

Base Materials for Printed Circuit Boards

PCB and Material Day

University of Applied Sciences
April 1st 2022, Darmstadt

h_da


ELEKONTA MAREK

isola

Adolf Sommer, ISOLA GmbH

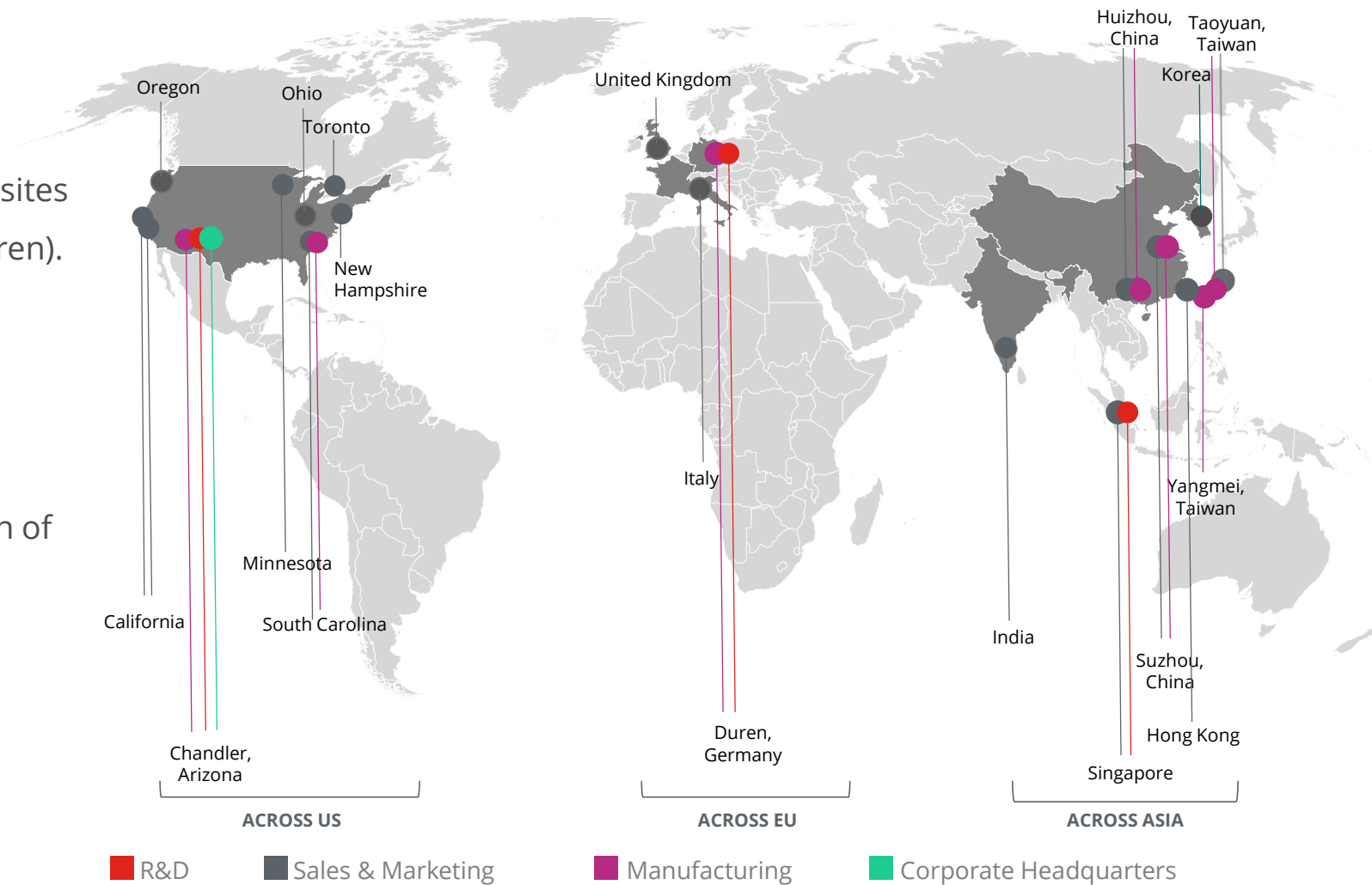
Content

1. Isola Introduction
2. Raw materials
3. Base material production
4. Material characteristics
5. Specification of base materials

> Isola – a short introduction

Isola is a Base Material manufacturer with production sites in the US, Asia and Europe (Düren).

We supply Base Materials, laminates and prepregs, to pcb shops for the production of printed circuit boards.



> Isola Market Segment



Wired Telecommunications & Computing

- Highest volumes still in High & Standard Loss sub-segments
- Highest growth in High Speed Digital (HSD) sub-segments
- HSD sub-segments range (Mid-Loss, Low Loss, Very Low Loss, Ultra Low Loss)
- Emerging requirements for even lower loss materials (i.e. to support wired 5G infrastructure)



Wireless Telecommunications - RF/Microwave

- Wireless telecommunications infrastructure equipment
- Mobile computing, smart devices, IoT, etc.
- Automotive ADAS: Advanced Driver Assistance Systems



Automotive Electrification & High Reliability

- Electric (EV) & Hybrid Electric (HEV) Vehicle electronics
- High power & high voltage requirements, driving
- Very demanding thermal reliability requirements
- Extreme CAF & electro-migration resistance requirements
- Thermal management solutions

> Product Portfolio

		Tachyon™100G Dk 3.02, Df 0.0021			
		I-Tera®MT40 Dk 3.45, Df 0.0031	Astra™MT77 Dk 3.00, Df 0.0017		
		IS550H Tg 200	TerraGreen® Dk 3.44, Df 0.0039	TerraGreen™ Dk 3.38 - 3.75 Df 0.0032 – 0.00,36	
		185HR Tg 180	I-Speed® Tg 180, Df 0.006	I-Tera®MT40 Dk 3.38 - 3.75 Df 0.0028 - 0.0035	
		370HR Tg 180	FR408HR Tg 190, Df 0.0092	IS680 AG Dk 3.00 – 3.45 Df 0.0020 – 0.0026	P25N Polyimide No Flow
DE104 Tg 135	IS400 Tg 150	IS415 Tg 190, Mid Dk/Df	IS680 Dk 2.80 – 3.45 Df 0.0025 – 0.0035	370HRNoflow	P95/P25 HB Polyimide
Standard Product	High Reliability FR4	High Reliability High Speed Digital	RF/Microwave	No/Low Flow Prepregs	Speciality Products
Low Tg	Mid Tg/High Tg	High Tg Low Dk/Df	Low Dk, Low Df	High Tg	High Tg
FR4 materials		HSD/ RF materials		Our products are RoHS/REACH compliant	

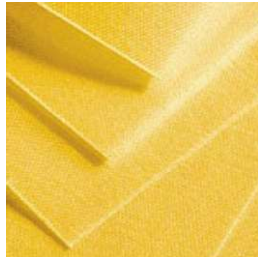
2. Raw Materials

2. Raw Materials



Prepreg

Glass fabric impregnated with B-stage resin.

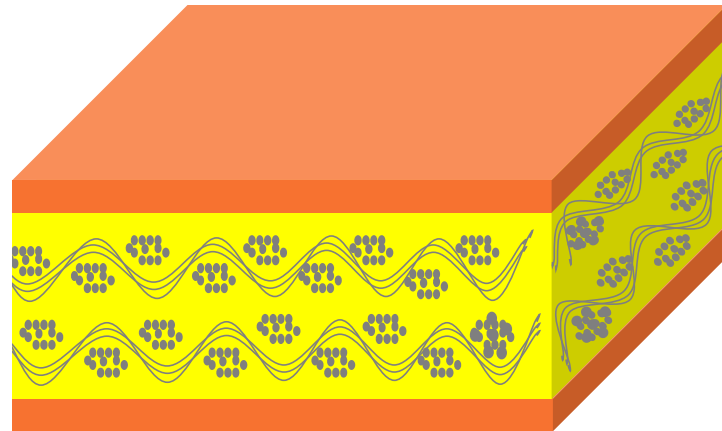


Laminate / core

Prepreg plies pressed between copper foil.



schematic illustration of a 2 ply laminate



2. Raw Materials



glass



copper



resin



Glass and **Copper** are

- finished raw materials
- ready for base material production



Resin is produced at Isola with
partially finished raw materials

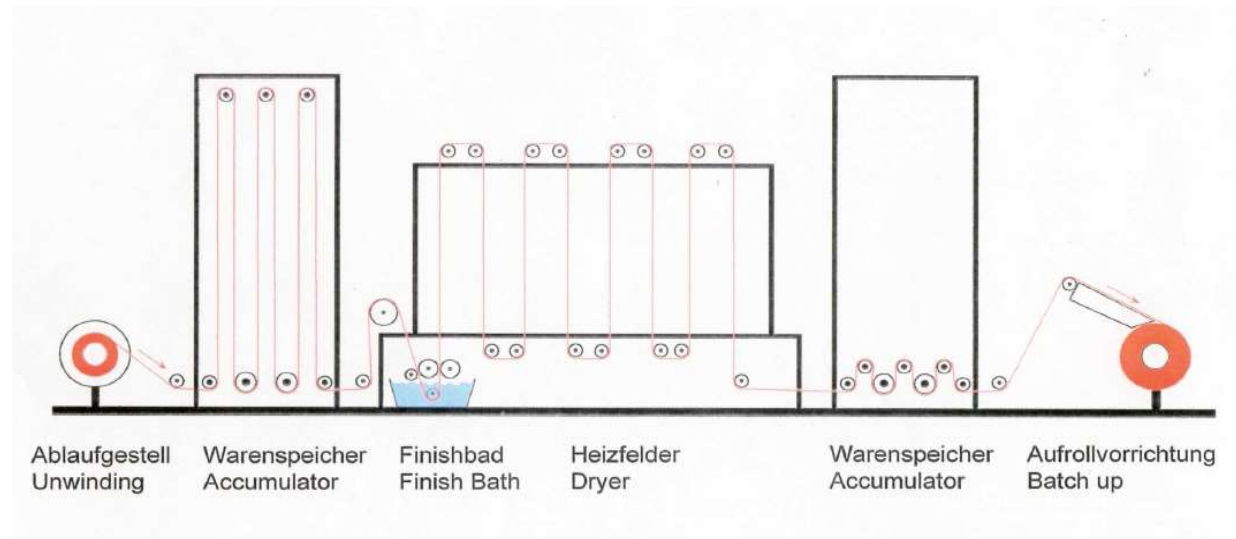
2. Raw Materials: Glass



Glass fabric production

- › Yarn production from a glass melt
- › Section beam production
- › Sizing of the yarn
- › Weaving of glass fabric on a loom
- › Thermal desizing
- › Application of finish

Finish application

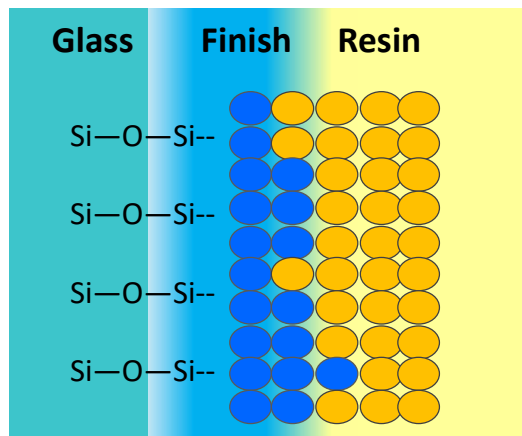
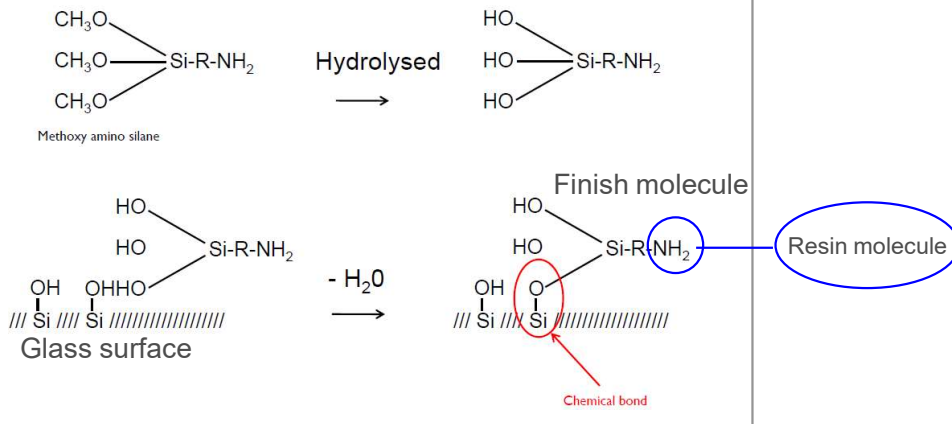


2. Raw Materials: Glass

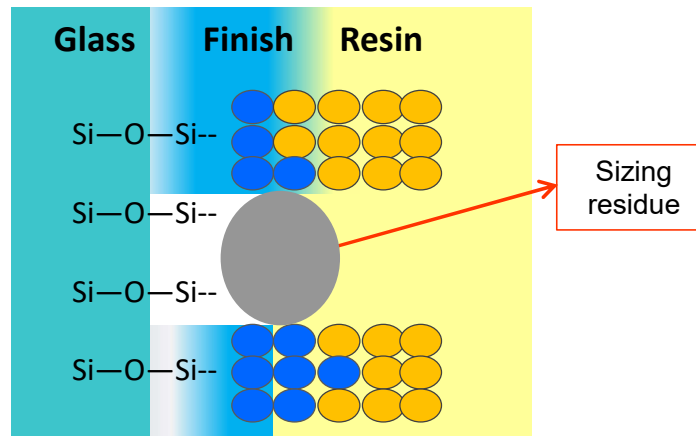
✓ Glass finish

- Finish is the adhesion promoter between glass and resin
- Chemical reaction of finish and glass surface
→ sizing must be removed completely before applying the finish
- Finish must be applied on complete glass surface

Finish mechanism



Complete wetting of the glass surface with finish and resin

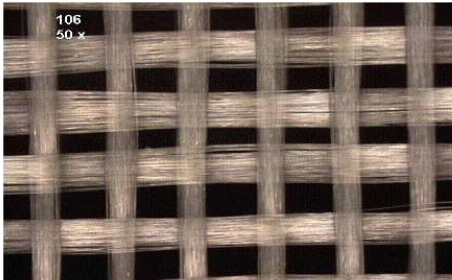


Residues of sizing on glass surface lead to non-wetted glass surface and voids

2. Raw Materials: Glass

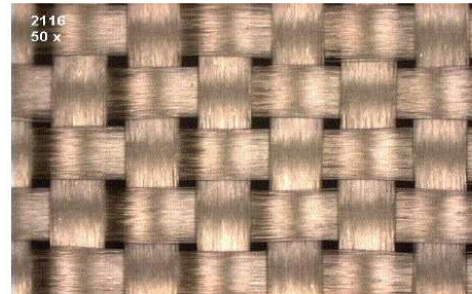


Glass fabric types



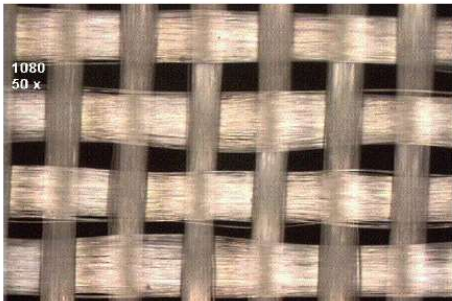
Type: 106

warp & weft : 22 x 22 (threads/cm)
thickness: 0.033 mm



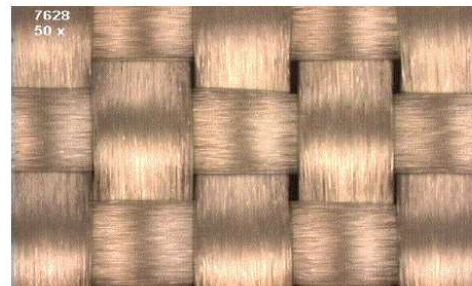
Type: 2116

warp & weft : 23,6 x 22,8 (threads/cm)
thickness: 0.094 mm



Type: 1080

warp & weft : 23,6 x 18,5 (threads/cm)
thickness: 0.053 mm



Type: 7628

warp & weft : 17,3 x 12,2 (threads/cm)
thickness: 0.173 mm

2. Raw Materials: Glass



Typical Glass Styles

Glass Style	Glass Thickness [mm]	Weight [g/m ²]	Threads per cm	Yarn
7628	0,170	203	17,3 x 12,2	EC9 68/EC9 68
2116	0,095	104	23,6 x 22,8	EC7 22/EC7 22
3313	0,084	81	23,6 x 24,4	EC6 16,5/EC6 16,5
2113	0,079	78	23,6 x 22,0	EC7 22/EC5 11
2125	0,090	87	15,7 x 15,4	EC7 22/EC9 34
1086	0,054	54	23,6 x 23,6	EC5 11/EC5 11
1080	0,050	47	23,6 x 18,5	EC5 11/EC5 11
1067	0,035	31	27,6 x 27,6	EC5 5,5/EC5 5,5
106	0,033	24	22,0 x 22,0	EC5 5,5/EC5 5,5
1037	0,027	23	27,6 x 28,7	EC4,5 4,1/EC4,5 4,1



IPC-4412 Specification for Finished Fabric Woven from „E“ Glass for Printed Boards



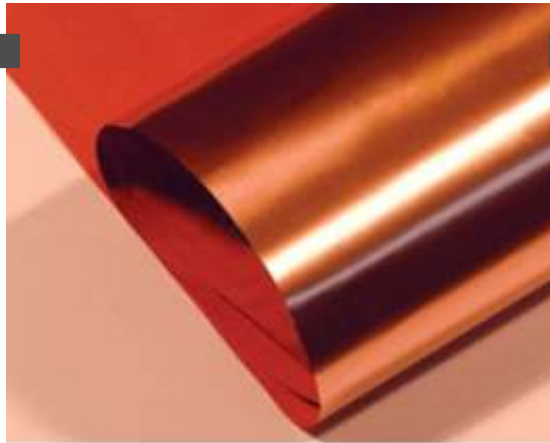
E **C** **5** **11** Designation of glass

E E-glass (glass type)
 C continuous (yarn type)
 5 filament diameter in µm
 11 yarn weight in tex (=g/km)

2. Raw Materials: Copper

Treated / Matte Side of the copper:

Inner side of the copper facing the prepreg. This copper treatment forms the rough surface into which the resin of the prepregs can interlock.



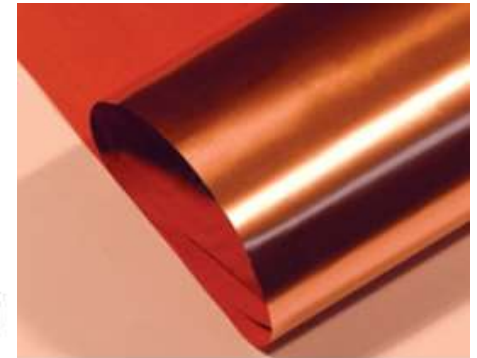
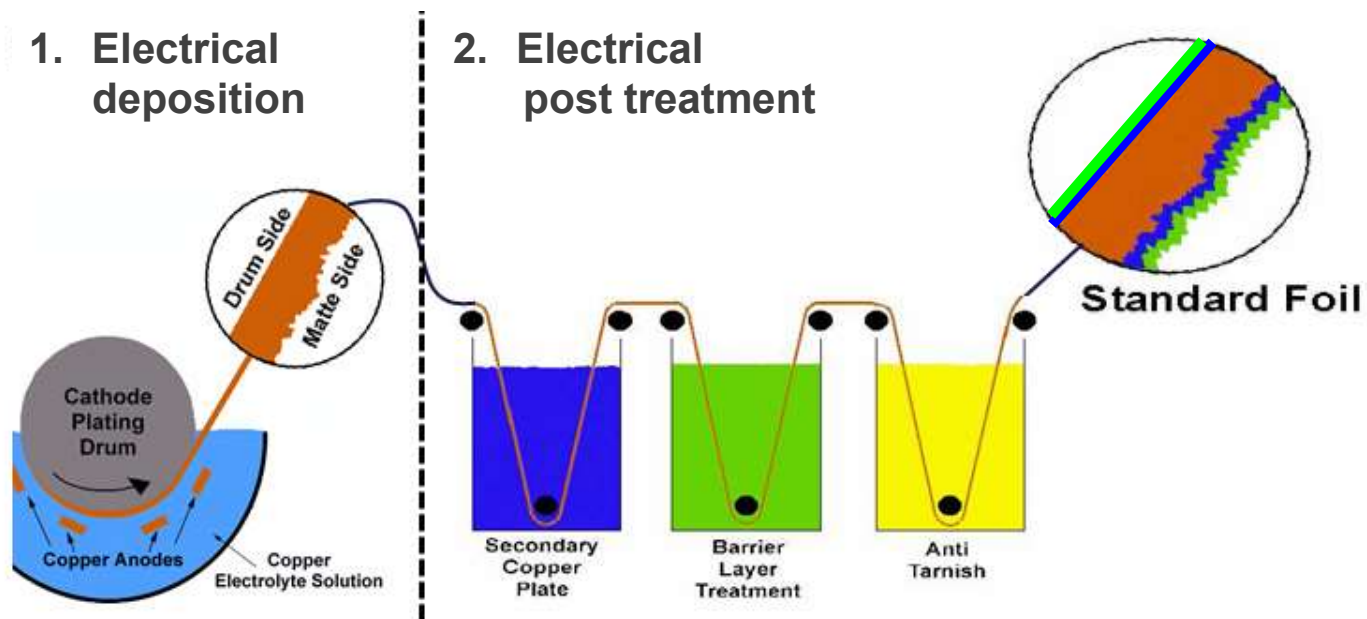
Drum / Shiny Side of the copper:

Outer side of the copper, ready for further processing in pcb production.

2. Raw Materials: Copper



ED copper: electrically deposited copper



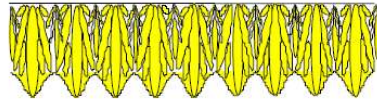
2. Raw Materials: Copper

cross section picture of a standard copper foil

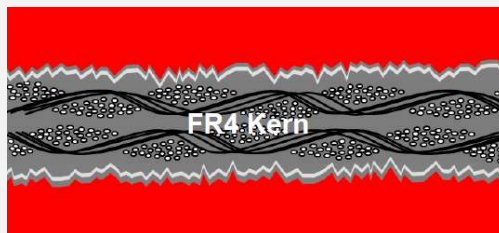
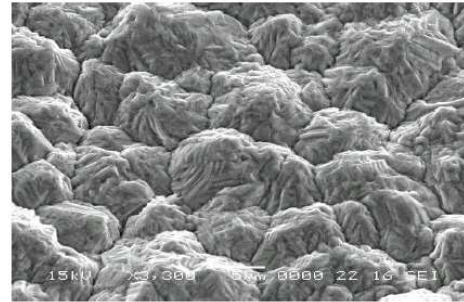
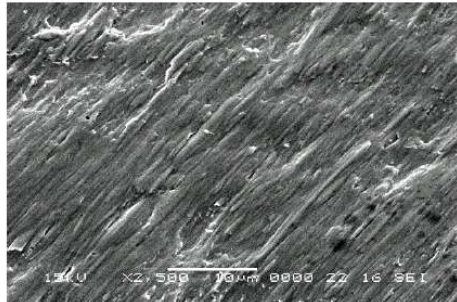


Shiny Side

Electrodeposited foil



Matte Side



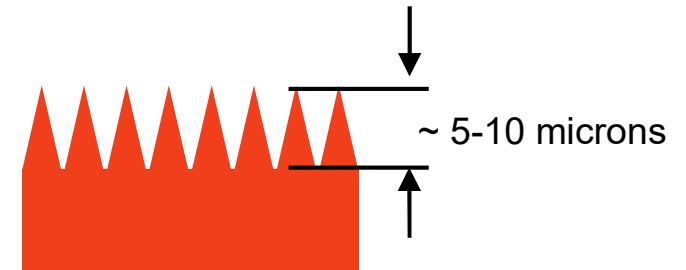
Shiny side

Matte side

FR4 Kern



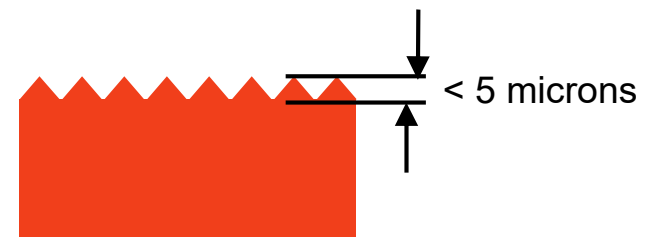
Standard Copper Profile



35 μm foil: 5 - 8 μm
70 μm foil: <10,2 μm
105 μm foil: 12 - 18 μm
210 μm foil: < 20 μm



Very Low Copper Profile



2. Raw Materials: Copper

> Typical copper thicknesses

Foil Designation	Common Industry Terminology	Metric	
		Area Weight (g/m ²)	Nominal Thickness (µm)
E	5 µm	45.1	5.1
Q	9 µm	75.9	8.5
T	12 µm	106.8	12.0
H	1/2 oz	152.5	17.1
M	3/4 oz	228.8	25.7
1	1 oz	305.0	34.3
2	2 oz	610.0	68.6
3	3 oz	915.0	102.9
4	4 oz	1220.0	137.2
5	5 oz	1525.0	171.5
6	6 oz	1830.0	205.7
7	7 oz	2135.0	240.0
10	10 oz	3050.0	342.9
14	14 oz	4270.0	480.1

IPC-4562 Table 1-1



Copper foil thicknesses and the corresponding Designations commonly used in the market.

oz	Thickness	remark
½ oz	18 µm	Standard foils
1 oz	35 µm	
2 oz	70 µm	
3 oz	105 µm	Heavy copper foils
6 oz	210 µm	
12 oz	400 µm	



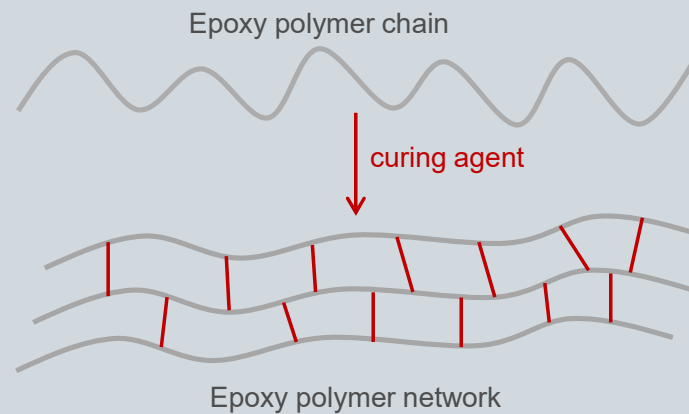
IPC-4562 Metal Foil for Printed Wirings Applications

2. Raw Materials: Resin

> Epoxy Resin

> Basic resin components

- > Epoxy polymer chain
- > Curing agent
- > Depending on resin system also fillers



2. Raw Materials: Resin

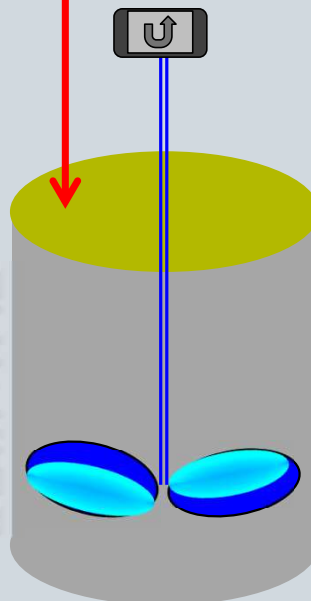


Resin Production at Isola:

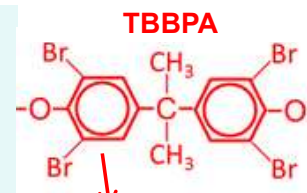
Insertion of the Flame Retardant: “bromination”

Basic epoxy resin
Tetrabromobisphenol A (TBBPA)
Catalyst
Solvent

REACTOR



BROMINATED
EPOXY RESIN



Basic epoxy resin molecule

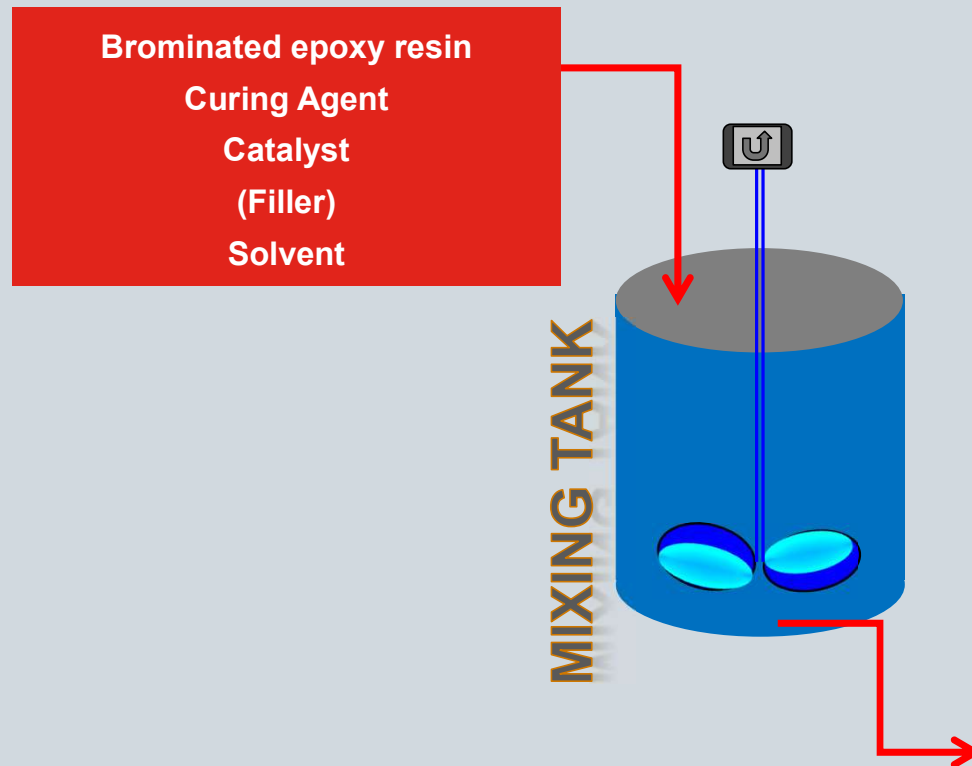


2. Raw Materials: Resin



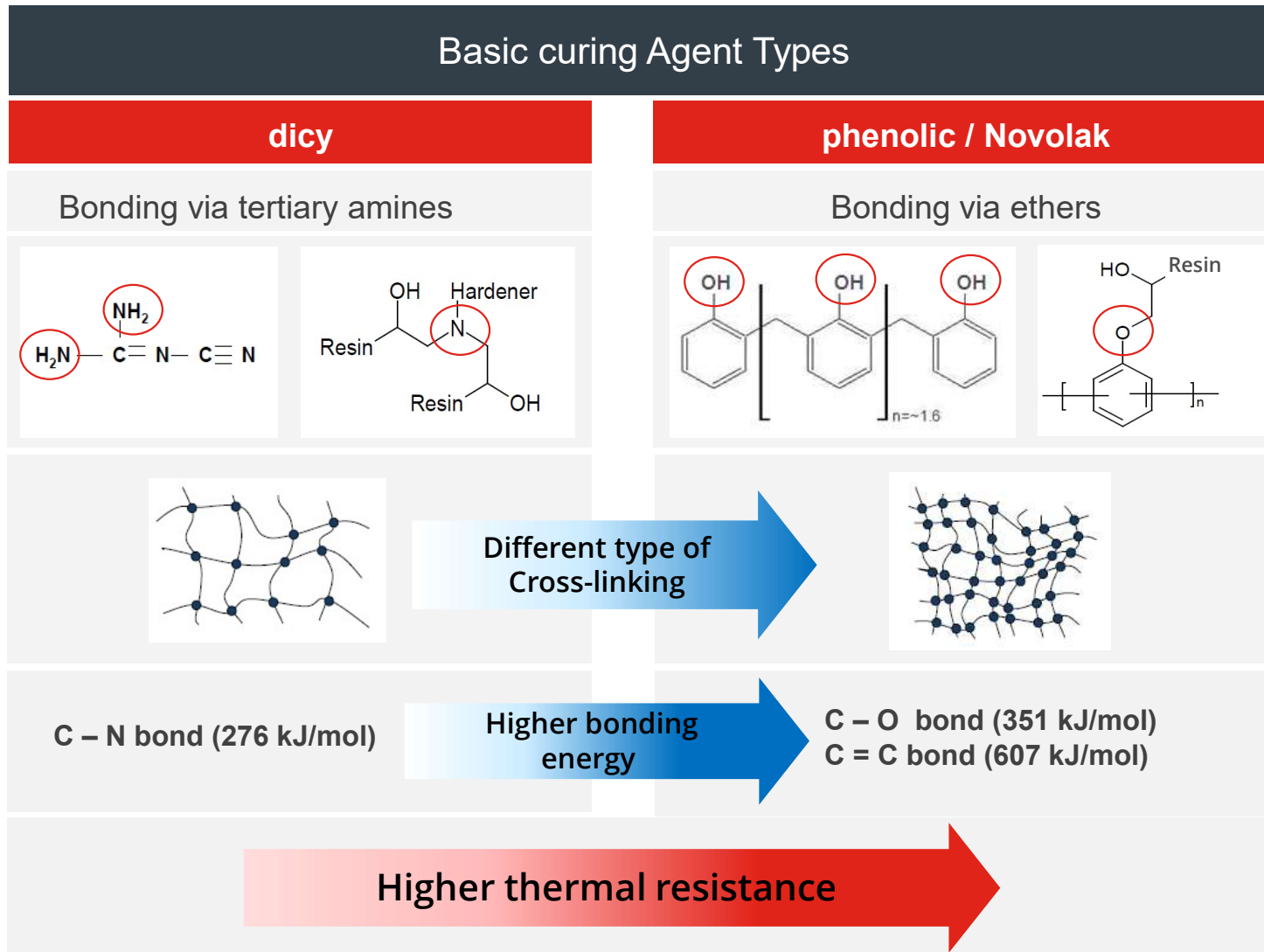
Resin Production at Isola:

Mixing of a certain resin product



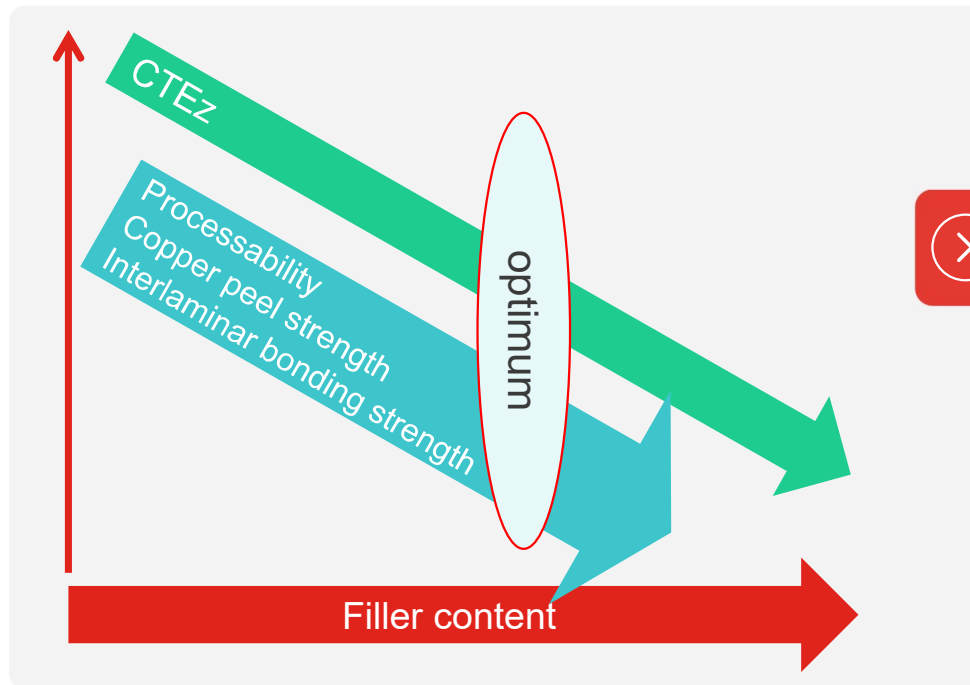
MATURE TANK

2. Raw Materials: Resin

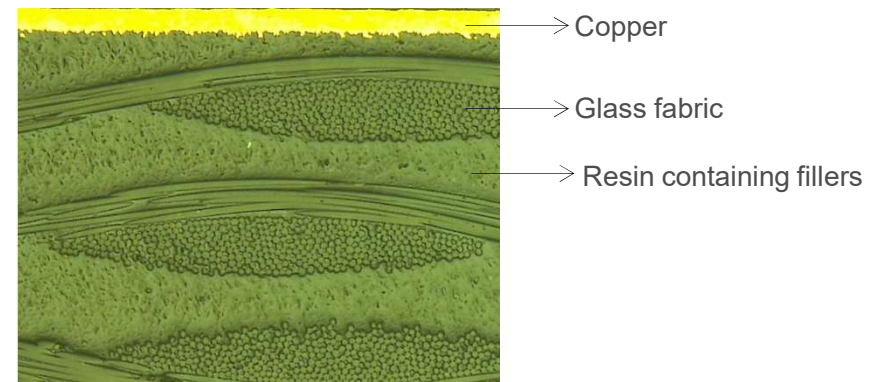


2. Raw Materials: Resin

> Fillers

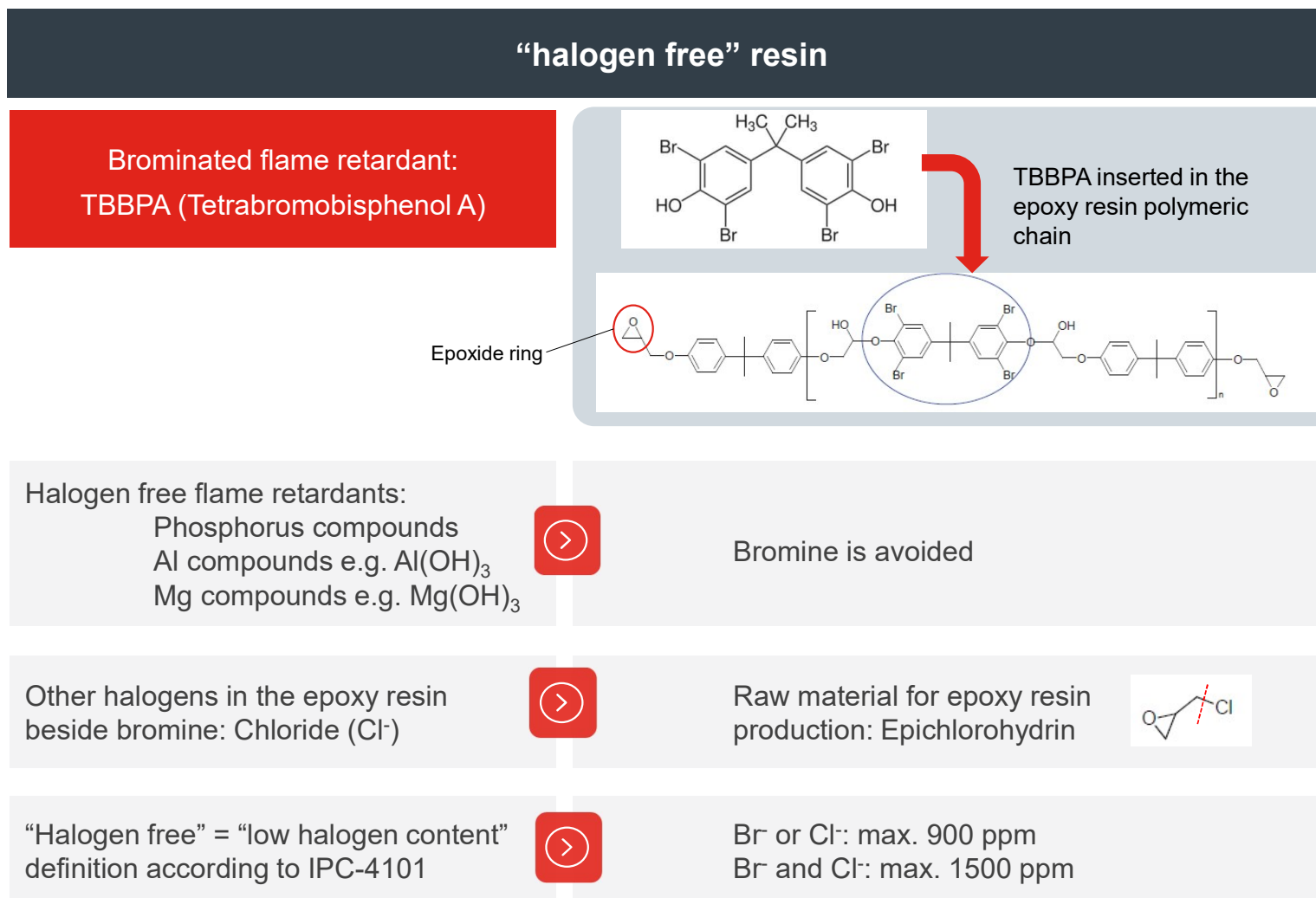


> Mineral,
inert,
insoluble,
non-conductive additives
that change the mechanical
properties of the resin.



Cross section picture of a laminate

2. Raw Materials: Resin



2. Raw Materials



With combining epoxy resin and glass we create **FR-4** materials



FR-4 defined by NEMA in 1968
(National Electrical Manufacturers Association)



FR = **F**lame **R**etardant



FR-4 is defined as :
a composite material, which is made of

- glass fabric and
- epoxy resin and
- is flame retardant

3. Base Material Production

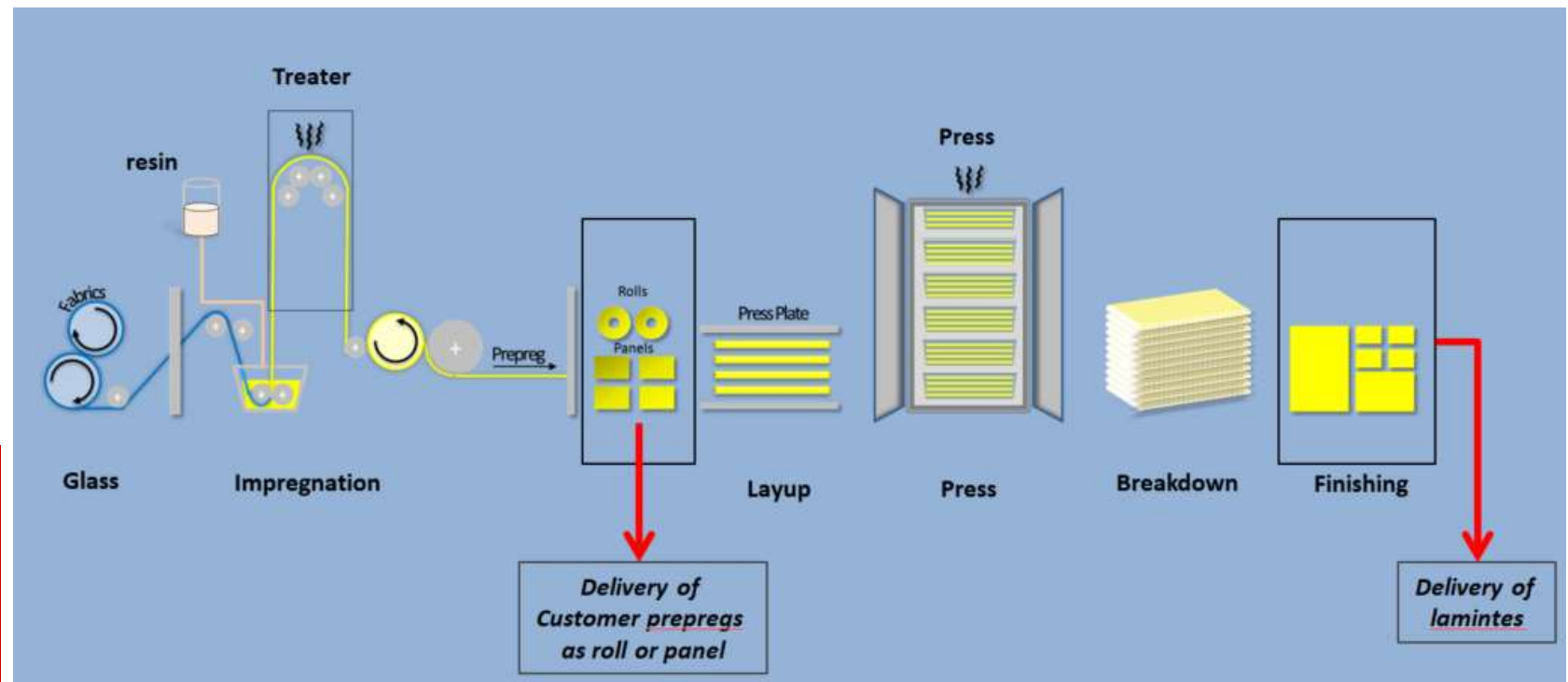
3. Base Material Production

Resin
Production:

Bromination
and Mixing

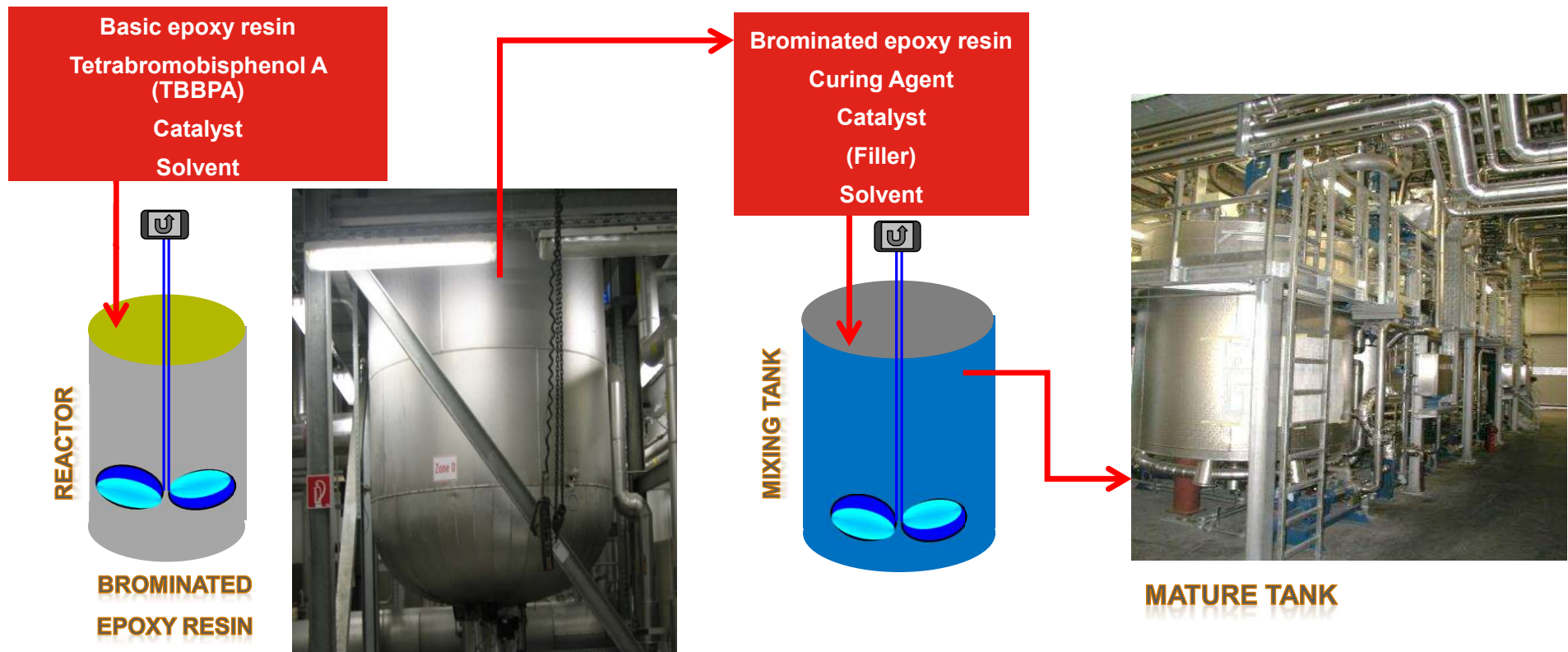
Prepreg
Production:
Impregnation

Laminate
Production:
Press

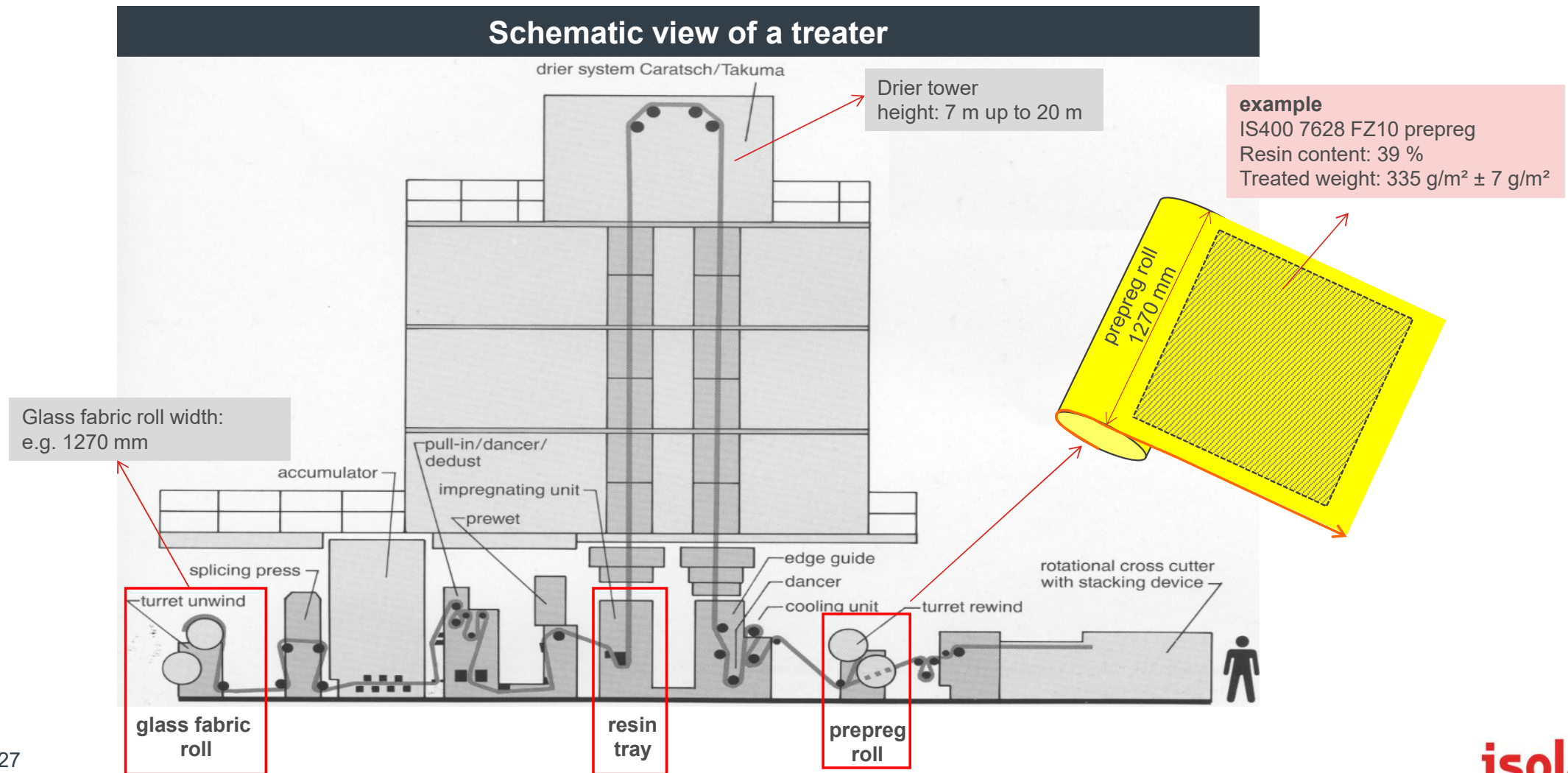


3. Base Material Production: Resin

Bromination and Mixing



3. Base Material Production: Prepreg



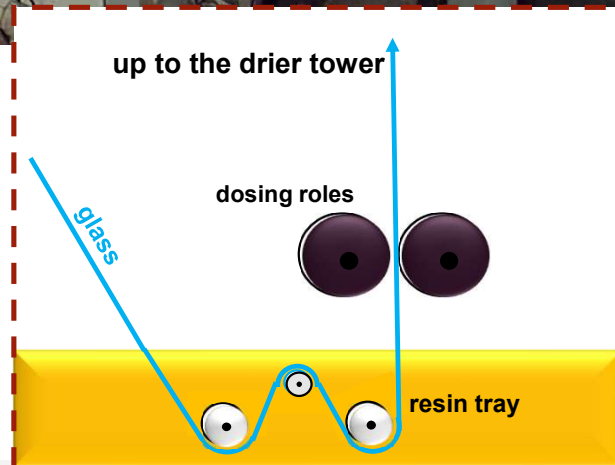
3. Base Material Production: Prepreg

Glass unwinding station



3. Base Material Production: Prepreg

Varnish tray and prepreg



B-Stage Prepreg

- Prepreg: Pre-impregnated
- B-stage: pre-cured resin

Ongoing testing:

- Weighing:
Amount of applied resin
- Minimum Viscosity and
Gelttime:
Rheology / Flow behavior
- Optical inspection

3. Base Material Production: Prepreg

Typical Prepreg Types, e.g. IS400

Glass Style	Type	Resin Content	Calculated Thickness After Pressing
106	FZ01	74%	0,057 mm
106	FZ98	79%	0,072 mm
1080	FZ01	65%	0,078 mm
1080	FZ99	69%	0,089 mm
1080	FZ96	71%	0,095 mm
1080	FZ98	72%	0,099 mm
2113	FZ01	57%	0,101 mm
2116	FZ01	53%	0,119 mm
2116	FZ98	57%	0,134 mm
2125	FZ01	55%	0,107 mm
7628	FZ01	46%	0,199 mm
7628	FZ98	51%	0,221 mm

3. Base Material Production: Laminates

Fully automated Lay-up stations



Lay-up of the appropriate number of prepregs between two copper sheets



3. Base Material Production: Laminates

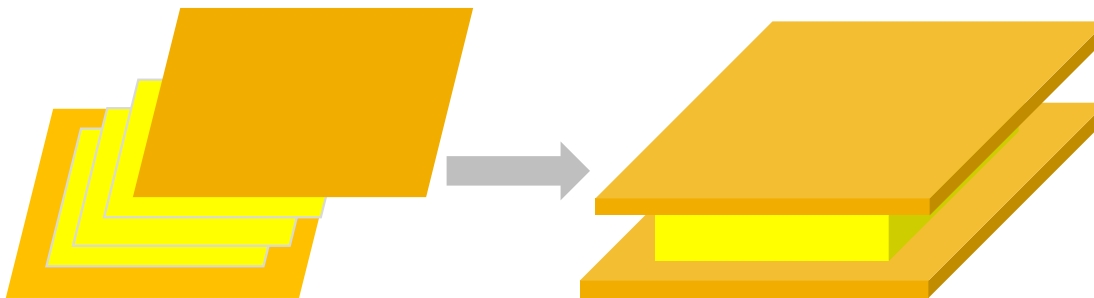
Laminate Presses



Typical sheet formats
that are pressed in the
laminate presses:

1225 mm x 925 mm

1225 mm x 1070 mm



Thickness tolerance:

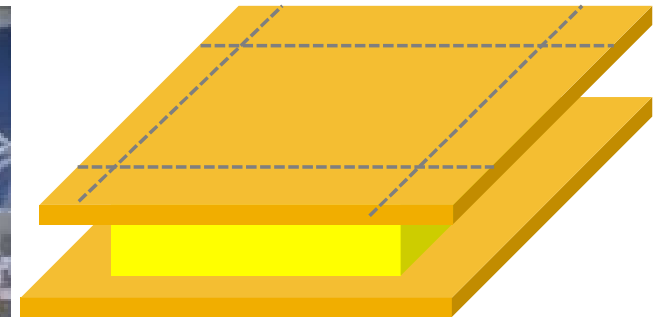
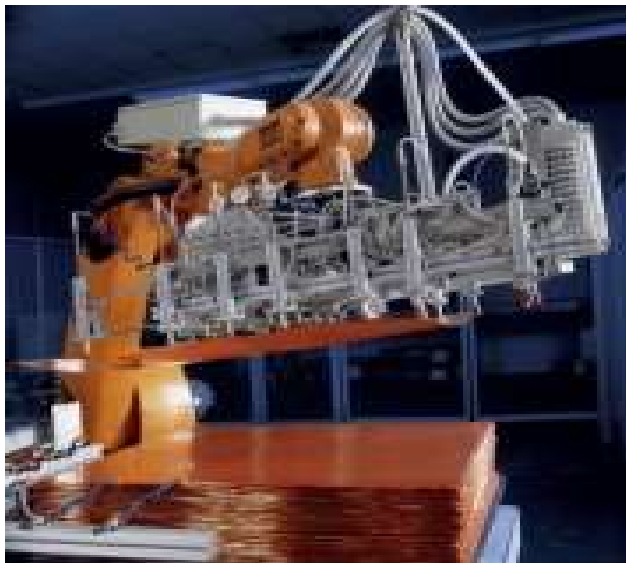
example

0,510 mm laminate

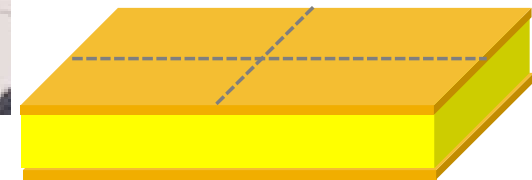
tolerance class B: $\pm 0,064$ mm

3. Base Material Production: Laminates

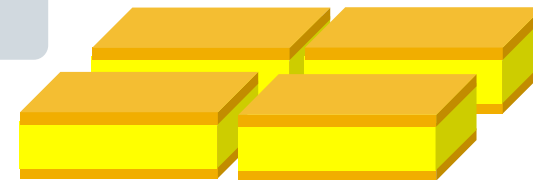
Laminate Finishing



Trimming



Cut to size



Typical panel formats that are cut for the pcb production:
610 mm x 460 mm
610 mm x 530 mm

3. Base Material Production: Laminates

Typical Laminate Types, e.g. IS400

Nominal thickness (substrate without copper)	Thickness Tolerance Class B*	Construction	
		Number of prepregs	Glass style
0,060 mm	±0,018	1x	106
0,100 mm	±0,018	1x	2116
0,125 mm	±0,025	1x	2116
0,200 mm	±0,038	1x	7628
0,200 mm	±0,038	2x	2116
0,360 mm	±0,050	2x	7628
0,510 mm	±0,064	3x	7628
0,710 mm	±0,064	4x	7628
...	...	...	...
1,500 mm	±0,130	8x	7628
...	...	...	...

Special 2 ply construction

* Acc. to IPC-4101 Specification for Base Materials for Rigid and Multilayer Printed Boards

4. Base Material characteristics

4. Base Material Characteristics



How to choose a base material?



Common base material categories:

- FR-4
- Tg: Standard Tg / Middle Tg / High Tg
- Standard / HF – material

Is that sufficient?



Material Data Sheet →

4. Base Material Characteristics

IS400 Specifications

Thermal characteristics

IS400 Data Sheet

Isola-Website

www.isola-group.com

Property		Typical Values			
				Units	Test Method
		Typical Value	Specification	Metric (English)	IPC-TM-650 (or as noted)
Glass Transition Temperature (Tg) by DSC		150	110-150	°C	2.4.25
Decomposition Temperature (Td) by TGA @ 5% weight loss		330	—	°C	ASTM D3850
T260		>60	—	Minutes	2.4.25
T288		>10	—	Minutes	2.4.25
CTE, Z-axis	A. Pre-Tg	50	AABUS	ppm/°C	2.4.24
	B. Post-Tg	250	—	—	
CTE, X-, Y-axes	A. Pre-Tg	13	AABUS	ppm/°C	2.4.24
	B. Post-Tg	14	—	—	
Z-axis Expansion (50-260°C)		3.30	AABUS	%	2.4.24
Thermal Conductivity		0.36	—	W/mK	ASTM D5930
Thermal Stress 10 sec @ 288°C (550.4°F)	A. Unetched	Pass	Pass Visual	Rating	2.4.13.1
	B. Etched	—	—	—	
Dk, Permittivity (Laminate & prepreg as laminated) Tested at 50% resin	A. @ 1 MHz (Fluid cell)	4.00	5.40	—	2.5.5.3
	B. @ 500 MHz (HP4291)	3.90	—	—	2.5.5.9
	C. @ 1 GHz (HP4291)	—	—	—	2.5.5.5
Df, Loss Tangent (Laminate & prepreg as laminated) Tested at 50% resin	A. @ 1 MHz (Fluid cell)	0.020	0.035	—	2.5.5.3
	B. @ 500 MHz (HP4291)	0.022	—	—	2.5.5.9
	C. @ 1 GHz (HP4291)	—	—	—	2.5.5.5
Volume Resistivity	A. 96/35/90	4.0x10 ⁹	1.0x10 ⁹	MΩ-cm	2.5.17.1
	B. After moisture resistance	—	—	—	
	C. At elevated temperature	7.0x10 ⁷	1.0x10 ⁹	—	
Surface Resistivity	A. 96/35/90	3.0x10 ⁹	1.0x10 ⁹	MΩ	2.5.17.1
	B. After moisture resistance	—	—	—	
	C. At elevated temperature	5.4x10 ⁹	1.0x10 ⁹	—	
Dielectric Breakdown		>50	—	kV	2.5.6
Arc Resistance		120	60	Seconds	2.5.1
Electric Strength (Laminate & prepreg as laminated)		48 (1100)	29 (736)	kV/mm (V/mil)	2.5.6.2
Comparative Tracking Index (CTI)		3 (175-249)	—	Class (Volts)	—
Peel Strength	A. Low profile copper foil and very low profile – all copper weights >17 microns B. Standard profile copper 1. After thermal stress 2. At 125°C (257°F) 3. After process solutions	1.05 (8.0)	0.70 (4.0)	—	2.4.8
		—	—	N/mm (lb/inch)	2.4.8.2
		1.45 (9.0)	1.05 (8.0)	—	2.4.8.3
		1.25 (8.0)	0.70 (4.0)	—	—
		1.45 (9.0)	0.80 (4.5)	—	—
Flexural Strength	A. Lengthwise direction	82,000	—	lb/inch ²	2.4.4
	B. Crosswise direction	66,600	—	—	
Tensile Strength	A. Lengthwise direction	51,213	—	lb/inch ²	—
	B. Crosswise direction	41,675	—	—	
Young's Modulus	A. Grain direction	3663	—	ksi	www
	B. Fill direction	3328	—	—	
Poisson's Ratio	A. Grain direction	0.183	—	—	xx
	B. Fill direction	0.151	—	—	
Moisture Absorption		0.18	0.8	%	2.6.2.1
Flammability (Laminate & prepreg as laminated)		V-0	V-0	Rating	UL 94
Max Operating Temperature		130	—	°C	—

The data, while believed to be accurate and based on analytical methods considered to be reliable, is for information purposes only. Any sales of these products will be governed by the terms and conditions of the agreement under which they are sold.

4. Base Material Characteristics

Thermal characteristics,
that should be considered when assessing
the thermal resistance of base materials:

T_g

Glass Transition Temperature

T_d

Decomposition Temperature

CTE

Coefficient of Thermal Expansion

T260 / T288

Time to Delamination

4. Base Material Characteristics

Glass Transition Temperature T_g

- The “glass transition” is the transition from a **hard, brittle, glassy** state to a **soft, ruby** state of the resin.
- The T_g value was for a long time used as the only thermal characteristic value to determine the thermal resistance of base materials.
Meanwhile we know, that it is NOT the only value that should be considered when choosing a base material. But the general rule that the **Continuous Operating Temperature of a device has to be app. 15°C – 20°C below the T_g value** of a base material is still valid (at least for Standard- and Mid- T_g -Products).
- The importance of T_g is reflected in the fact that it is still used to classify base materials.
 - Standard T_g materials: T_g 130°C/135°C
 - Mid T_g materials: T_g 150°C
 - High T_g materials: T_g 170°C/180°C
- The T_g value is so important because **material properties change significantly** during the glass transition.

4. Base Material Characteristics

Glass Transition Temperature T_g : effect on pcb

Properties that change significantly during glass transition	Effects on the material	Possible effects on the PCB
Specific heat capacity	The amount of heat absorbed changes	-
Extension	The expansion of resin increases strongly compared to the expansion of copper: CTEz of resin* below T_g 70 ppm/K CTEz of resin above T_g 250 ppm/K CTE of copper 16,5 ppm/K	Cracks in the copper barrel Detachment of inner layers Delaminations
E-Module	The epoxy resin becomes soft and elastic	Delaminations between the layers Delaminations due to additional mechanical stress



Prolonged temperatures around / above T_g during manufacturing or operation of the PCB can lead to failures.

4. Base Material Characteristics

Glass Transition Temperature T_g

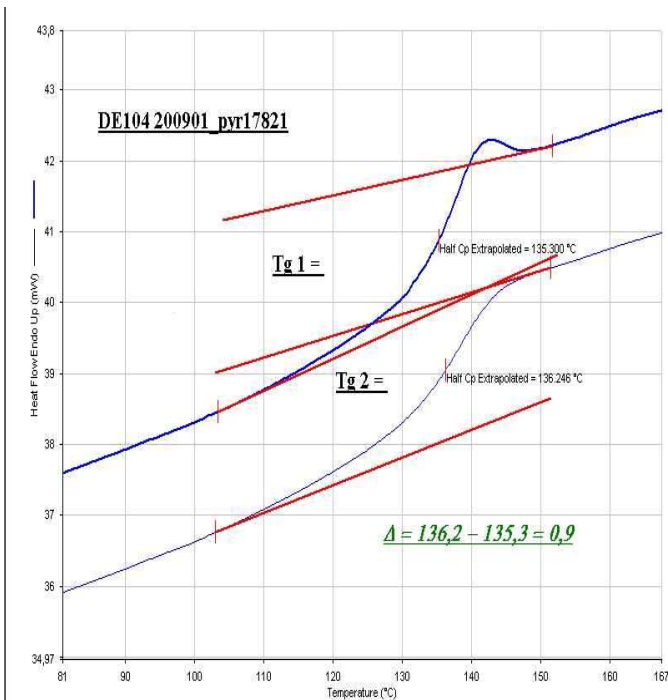


As three different material properties change significantly during the glass transition we have the ability to use three different test methods to measure the glass transition. These test methods can provide different test results.

- › Change of specific heat capacity DSC (Differential Scanning Calorimetry)
- › Change of elongation: TMA (Thermomechanical Analysis)
- › Change of E-Module: DMA (Dynamic Mechanical Analysis)

4. Base Material Characteristics

Glass Transition Temperature T_g



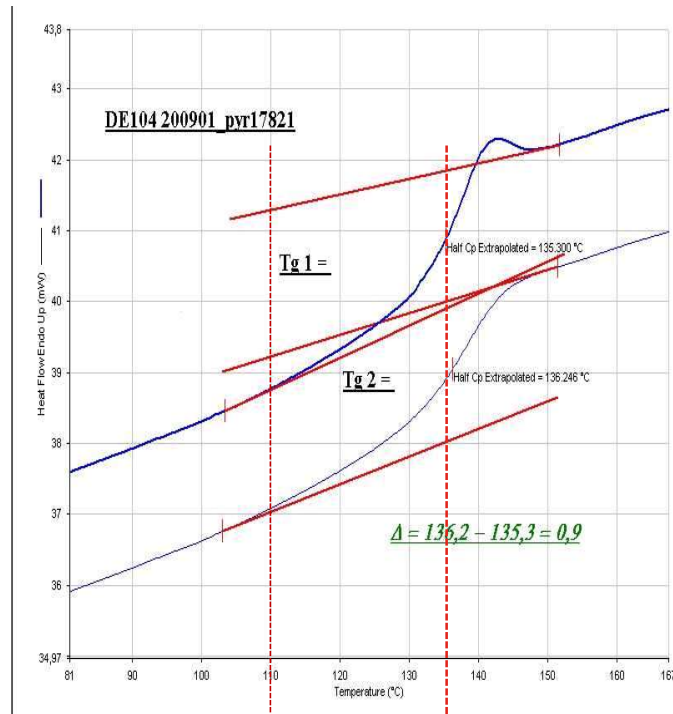
DSC – typical measurement curve

DSC – differential scanning calorimetry

- Change of heat flow as a function of temperature is measured
- Two successive measurements that allow a delta T_g to be recorded: $Tg2 - Tg1 = \text{delta } T_g$
- delta T_g could give indication on moisture or under-curing
- Dicy cured system: typical delta $T_g \sim 3^\circ\text{C}$
- Novolak cured systems: typical delta $T_g \sim 5^\circ\text{C}$

4. Base Material Characteristics

Glass Transition Temperature T_g : possible effects/failures on pcb due to high temperatures



Beginning of
glass transition ~ 110 °C

$T_g \sim 135 \text{ °C}$

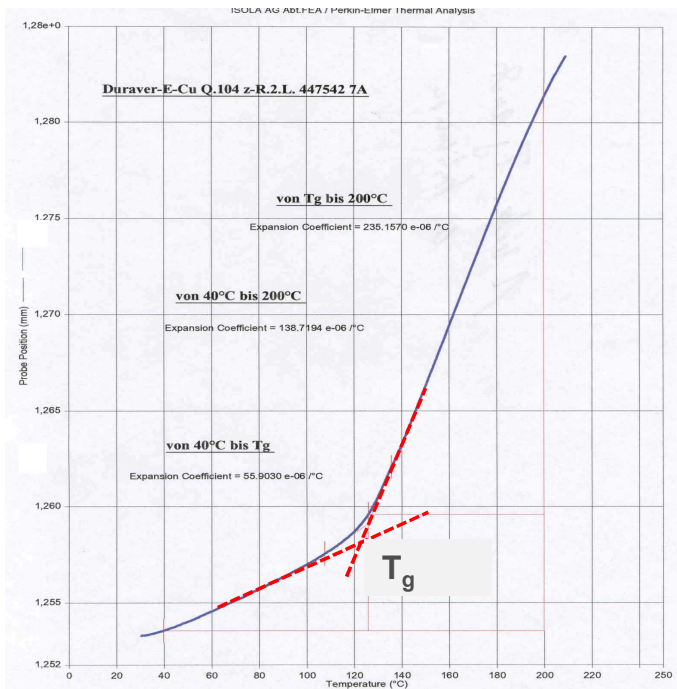
DSC – differential scanning calorimetry

Example to illustrate the continuous operating temperature rule

- Standard T_g material: $T_g \sim 135 \text{ °C}$
- Beginning of glass transition/glass transition effects: $\sim 110 \text{ °C}$
- Continuous operating temperature $15 - 20 \text{ °C} < T_g$
- This rule has its limitation when coming to $T_{op} > 150 \text{ °C}$

4. Base Material Characteristics

Glass Transition Temperature T_g : possible effects/failures on pcb due to high temperatures



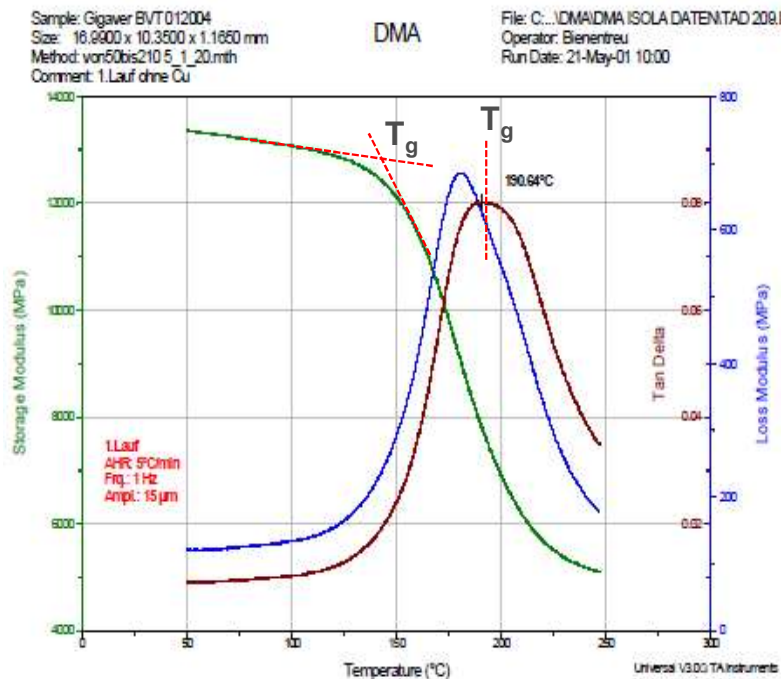
TMA – typical measurement curve in z-direction

TMA – Thermomechanical Analysis

- sample height as a function of temperature is measured

4. Base Material Characteristics

Glass Transition Temperature T_g : possible effects/failures on pcb due to high temperatures



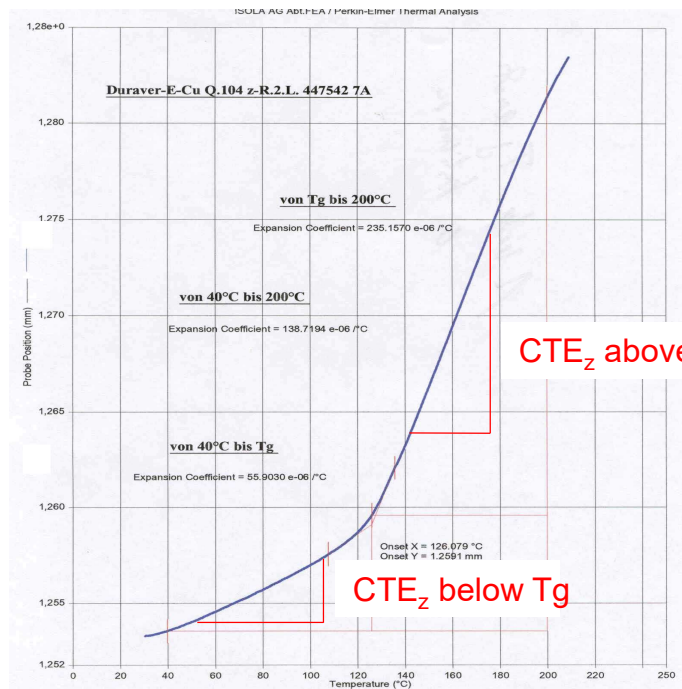
DMA – typical measurement curve

DMA – Dynamic mechanical analysis

- viscoelastic behaviour as a function of temperature is measured
- the **storage modulus** and the **loss modulus** are measured
tan delta is calculated from these values
- T_g is not defined;
even not within this one measurement where it can be the
turning point of the storage modulus or
the maximum of the tan delta

4. Base Material Characteristics

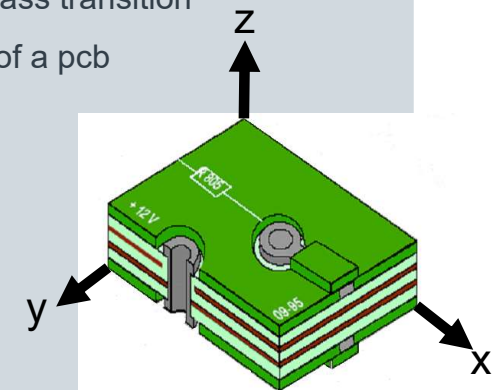
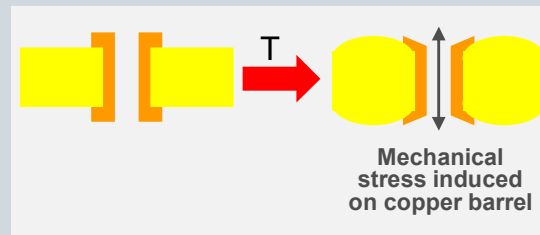
Coefficient of Thermal Expansion CTE



TMA – typical measurement curve

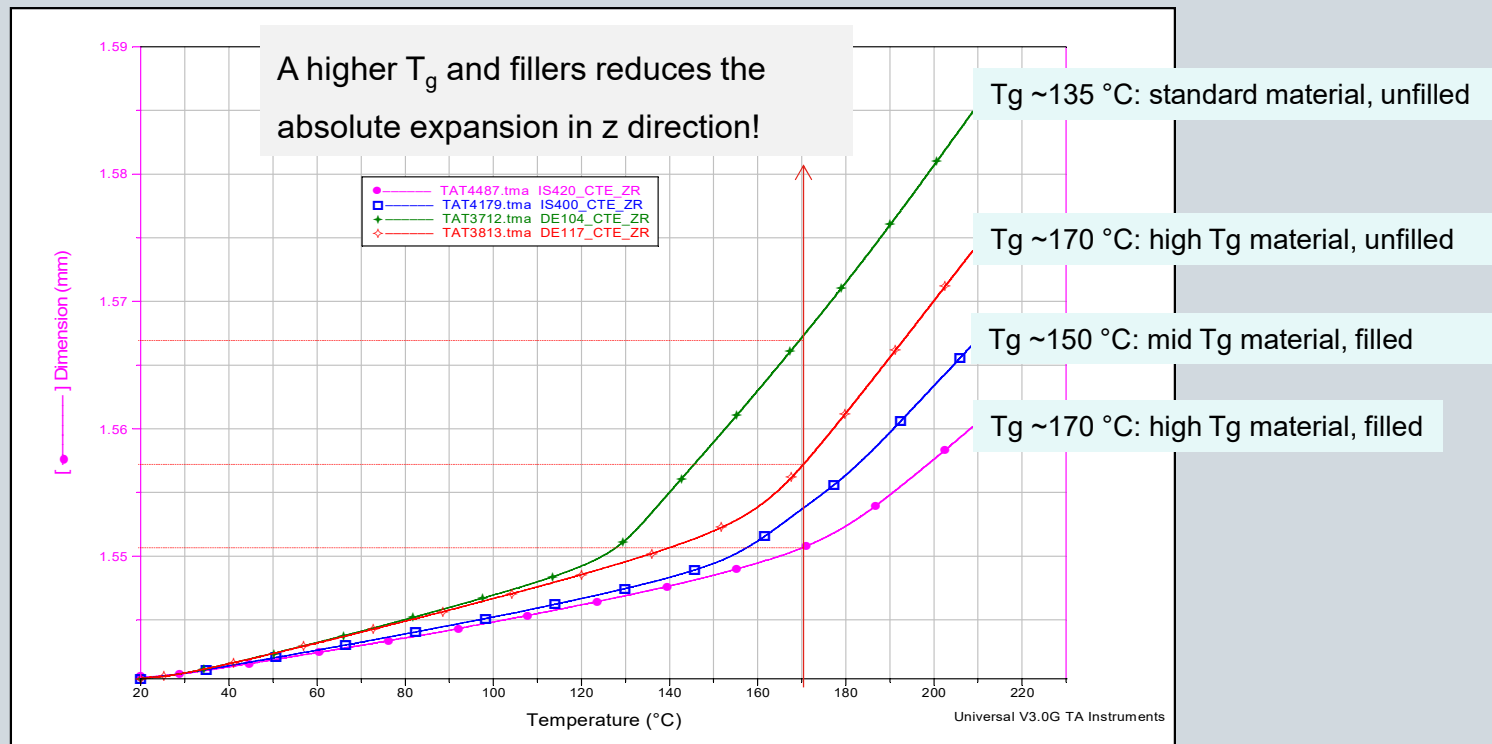
TMA – Thermomechanical Analysis

- sample height as a function of temperature is measured
- xy-direction is stabilized by the glass reinforcement
- z-direction is mainly affected by resin behaviour
 - CTE_z is sharply increasing during glass transition
 - CTE_z is significant for the reliability of a pcb



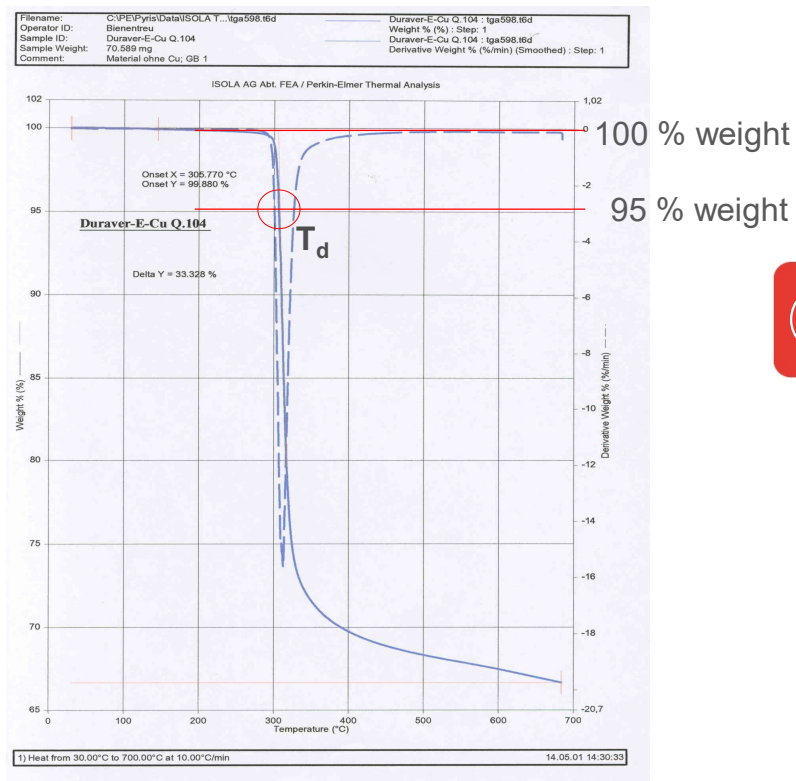
4. Base Material Characteristics

Coefficient of Thermal Expansion CTE



4. Base Material Characteristics

Decomposition Temperature T_d

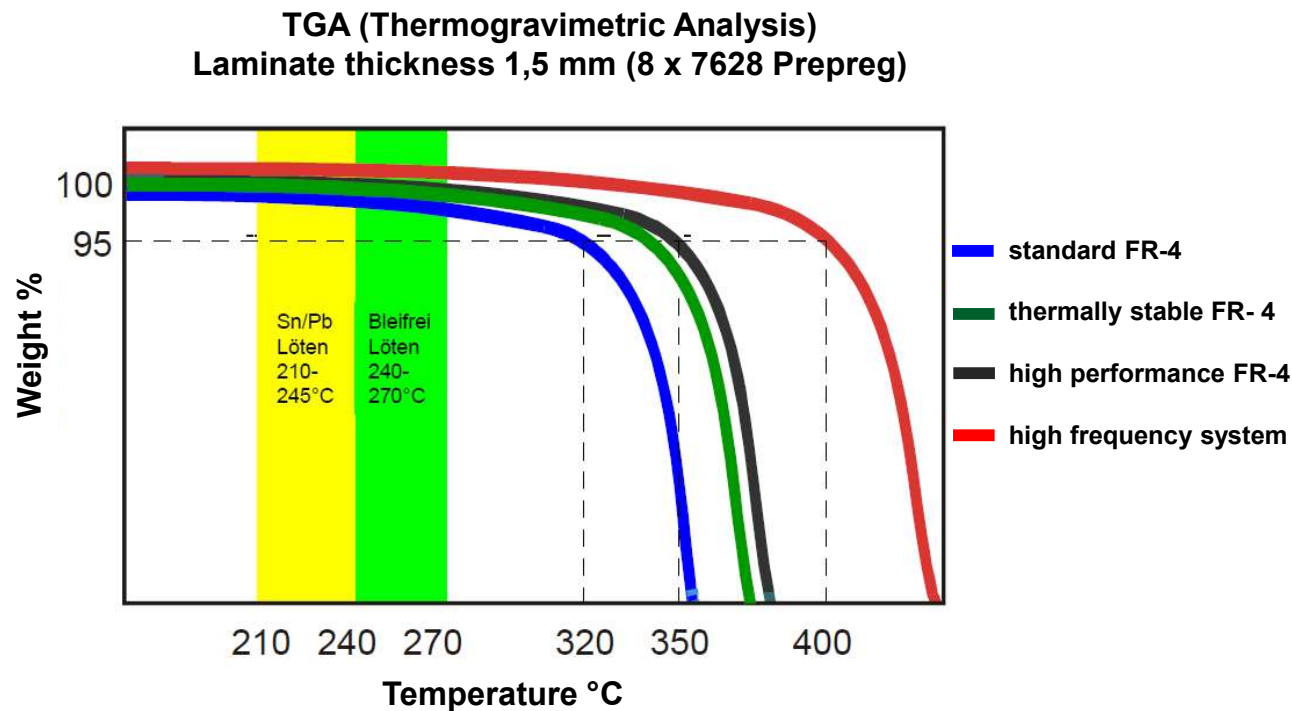


TGA – Thermogravimetric Analysis

- Weight of a sample is measured as a function of temperature
- According to IPC-4101 T_d is defined as the temperature at 5% loss of weight

4. Base Material Characteristics

Decomposition Temperature T_d



- Decomposition of the resin is irreversible.
- Thermal processes during pcb production and assembly can lead to delamination or even decomposition of the material.
- Thermal load during pcb production:
 - Multiple pressing cycles?
 - Lead free soldering?
 - Number of soldering cycles?
 - Hand soldering?
 - pcb construction (copper thickness, layer count, ...)?→ Is a thermal resistant material needed?

4. Base Material Characteristics

Lead-Free Soldering of standard base materials

Standard FR-4 materials (Tg 135°C, unfilled) are only partially suitable for lead-free soldering

→ Strongly dependent on the layout / construction of the pcb!

→ Always consider the overall heat transfer and heat absorption!



Uncladded laminate 1,5 mm thickness (7x7628 prepreg)

4. Base Material Characteristics

Lead-Free Soldering of standard base materials

> Cross section example of a standard Tg135 material after lead free reflow soldering (peak temperature 260°C)

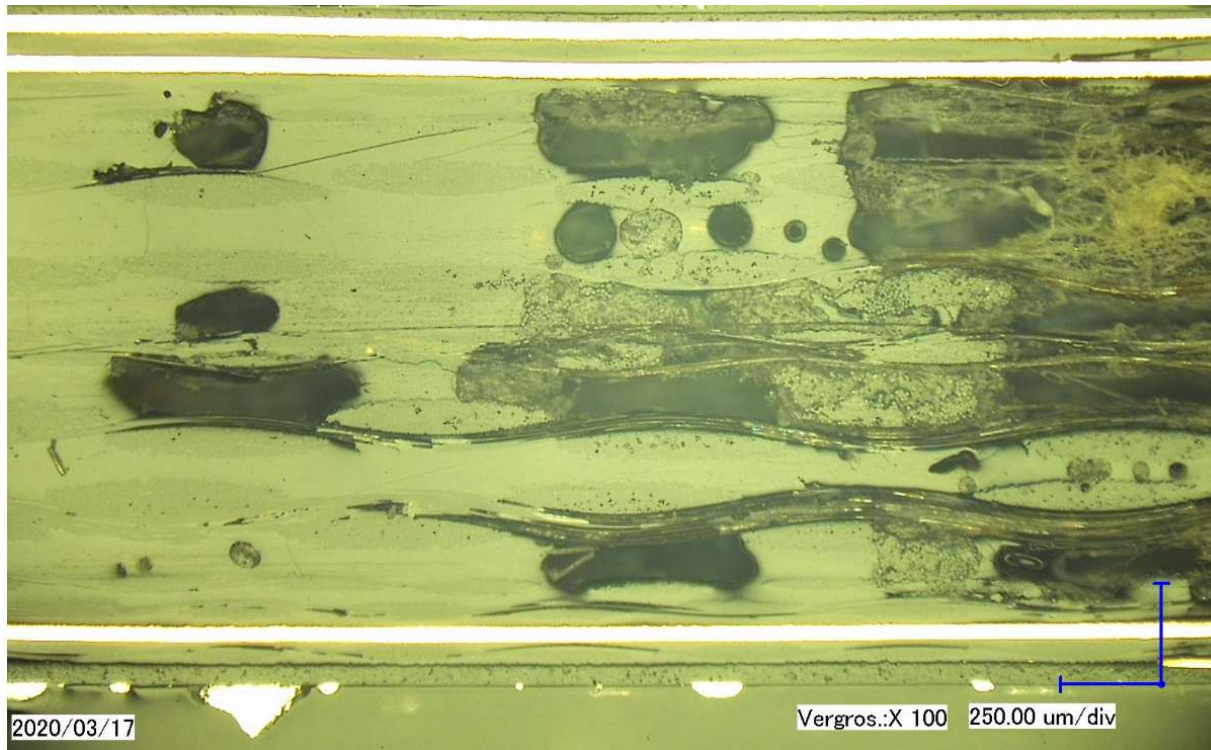


Areas within resin showing decomposition of the material

4. Base Material Characteristics

Lead-Free Soldering of standard base materials

➤ Cross section example of an standard Tg135 material after lead free reflow soldering (Peak temperature 260°C)

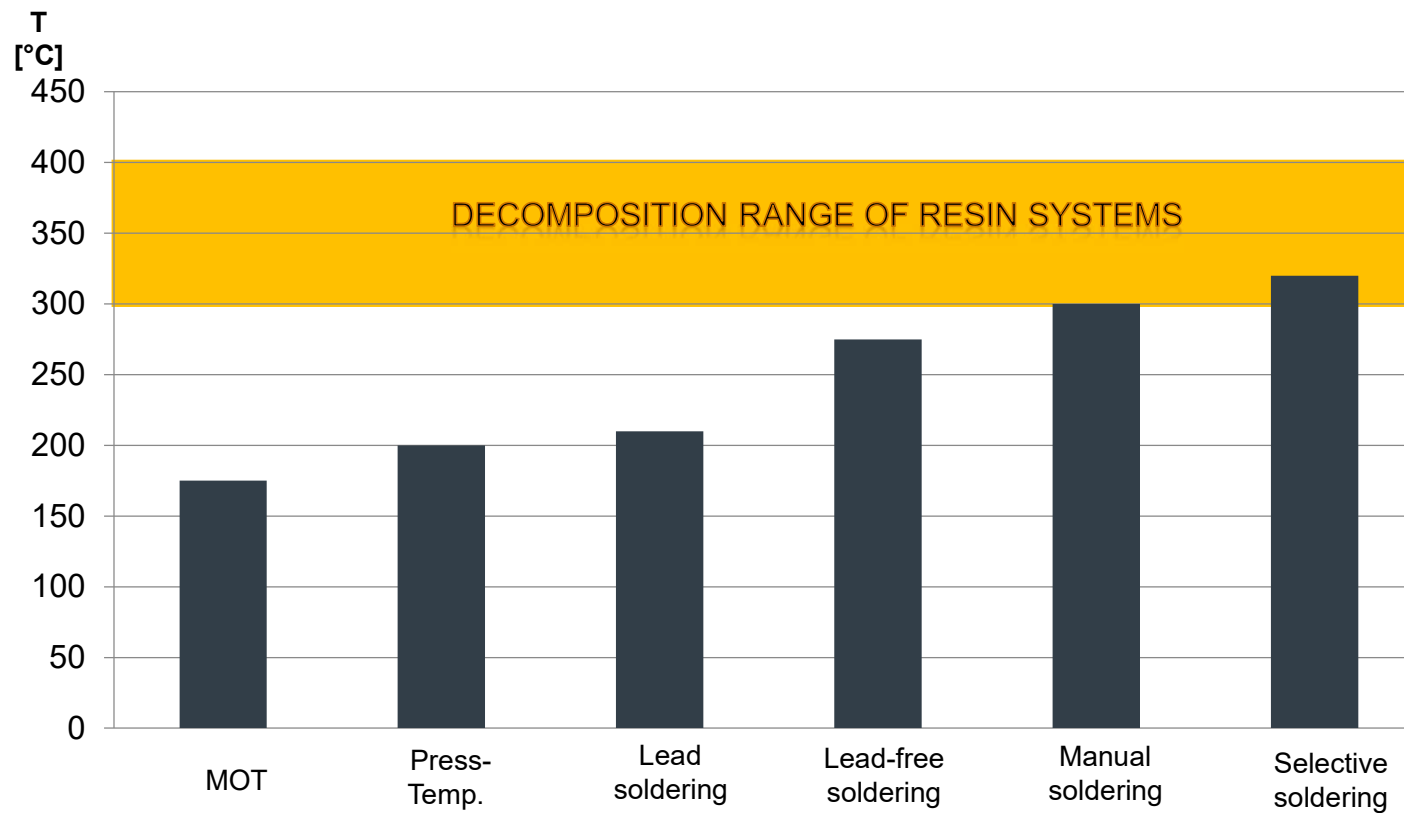


Extreme resin
Decomposition

→ Standard materials
Tg135 / dicy cured
are not recommended
for lead free soldering

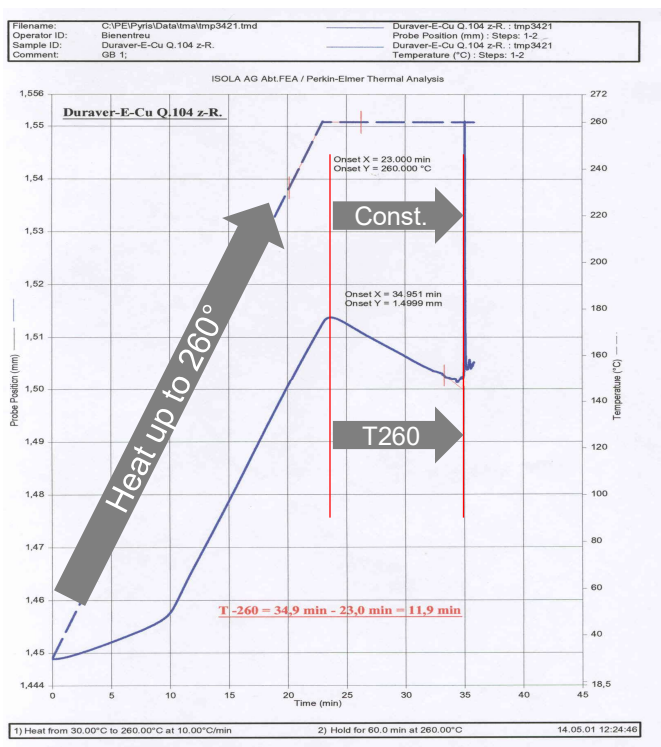
4. Base Material Characteristics

Decomposition Temperature Td



4. Base Material Characteristics

Time to Delamination T260/T288



TMA – typical measurement curve



Time to delamination after reaching test temperature is measured in the TMA.

In accordance with leaded and lead free soldering
Time to Delamination is measured at 260°C or 288°C.

4. Base Material Characteristics

Please note and remember



Data sheet values are

- › Dependant on resin/glass ratio
→ laminate construction dependant
- › Valid for laminates but not necessary for a pcb construction
- › Dependant on measurement method and measurement conditions
- › Are NOT fixed values,
but always “only” typical values
that initially only apply to the base material

4. Base Material Characteristics

Please note and remember



Thermal resistance/ reliability

of an assembly is not only influenced by the base material:

pcb layout:

- pcb thickness and layer construction (resin / glass relation)
- Inner layer copper thickness / percentage / distribution
- Drill diameter / aspect ratio

pcb production:

- Hole wall smoothness / desmear process
- Plating thickness
- Ductility of plated barrel copper

and all other process steps!

4. Base Material Characteristics



Base material **factors** influencing High Frequency signal transmission:

4. Base Material Characteristics

Resin

Base Material properties

- Dk: dielectric constant / ϵ_r / permittivity
- Df: dielectric loss factor / $\tan \delta$

IS400 Specifications

electrical characteristics

Property		Typical Values			
				Units	Test Method
		Typical Value	Specification	Metric (English)	IPC-TM-650 (or as noted)
Glass Transition Temperature (Tg) by DSC		150	110-150	°C	2.4.25
Decomposition Temperature (Td) by TGA @ 5% weight loss		330	—	°C	ASTM D3850
T260		>60	—	Minutes	2.4.25
T288		>10	—	Minutes	2.4.25
CTE, Z-axis	A. Pre-Tg	50	AABUS	ppm/°C	2.4.24
	B. Post-Tg	250	—		
CTE, X-, Y-axis	A. Pre-Tg	13	AABUS	ppm/°C	2.4.24
	B. Post-Tg	14	—		
Z-axis Expansion (50-260°C)		3.30	AABUS	%	2.4.24
Thermal Conductivity		0.36	—	W/mK	ASTM D5930
Thermal Stress 10 sec @ 288°C (550.4°F)	A. Unetched B. Etched	Pass	Pass Visual	Rating	2.4.13.1
Dk, Permittivity (Laminate & prepreg as laminated) Tested at 50% resin	A. @ 1 MHz (Fluid cell)	4.00	5.40	—	2.5.5.3
	B. @ 500 MHz (HP4291)	3.90	—		2.5.5.9
	C. @ 1 GHz (HP4291)	—	—		2.5.5.5
Df, Loss Tangent (Laminate & prepreg as laminated) Tested at 50% resin	A. @ 1 MHz (Fluid cell)	0.020	0.035	—	2.5.5.3
	B. @ 500 MHz (HP4291)	0.022	—		2.5.5.9
	C. @ 1 GHz (HP4291)	—	—		2.5.5.5
Volume Resistivity	A. 96/35/90	4.0x10 ⁹	1.0x10 ⁹	MΩ-cm	2.5.17.1
	B. After moisture resistance	—	—		
	C. At elevated temperature	7.0x10 ⁹	1.0x10 ⁹		
Surface Resistivity	A. 96/35/90	3.0x10 ⁹	1.0x10 ⁹	MΩ	2.5.17.1
	B. After moisture resistance	—	—		
	C. At elevated temperature	5.4x10 ⁹	1.0x10 ⁹		
Dielectric Breakdown		>50	—	kV	2.5.6
Arc Resistance		120	60	Seconds	2.5.1

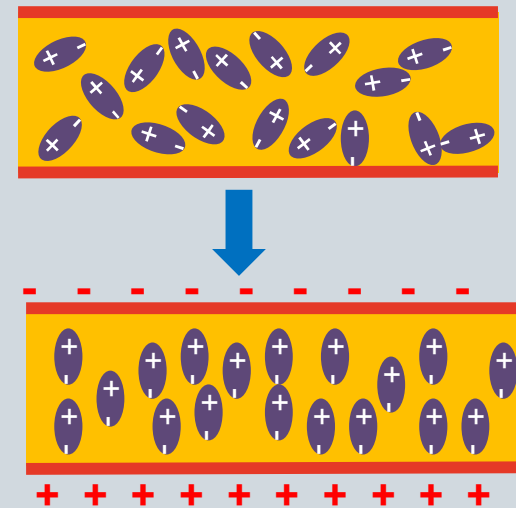
4. Base Material Characteristics

Resin: D_k



The application of an electric field causes polarization of charged particles in the base material and the formation of a polarization field, which acts against the applied field.

- D_k directly affects impedance
- Depends on the resin/glass ratio and frequency.
The higher the frequency, the lower the D_k values.



4. Base Material Characteristics

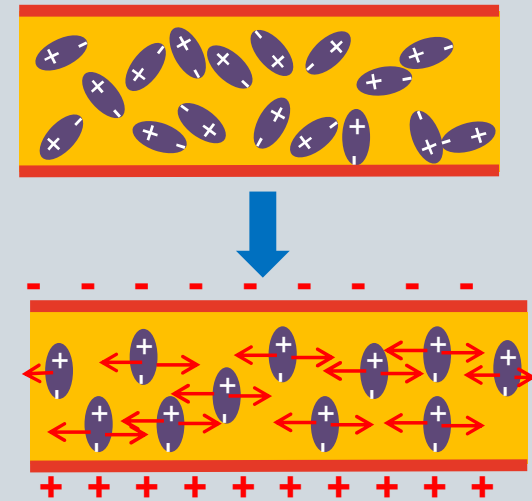
Resin: D_f



Particles are set into oscillation and generate friction heat.

Electrical energy is lost in form of heat, which leads to signal losses.

The higher the frequency, the higher the D_f values.

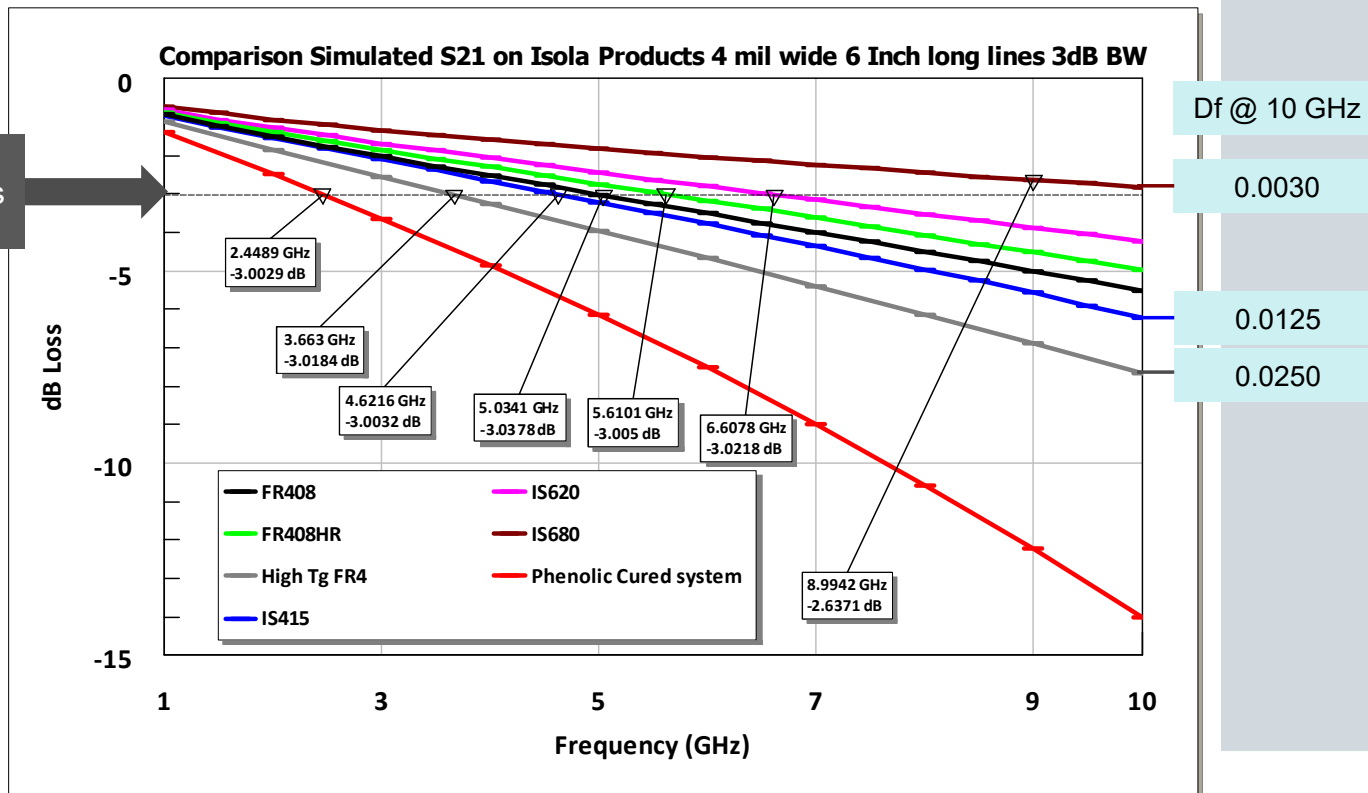


4. Base Material Characteristics

Resin: D_f



3 dB = 50 % signal loss



Resin: D_k and D_f

FR-4 materials:

construction overview with different laminates / prepregs and the according D_k/D_f values at certain frequencies



e.g. 370HR

High Tg material, epoxy resin

Core Data

Constructions	Resin Content %	Offering	Thickness (inch)	Thickness (mm)	Dielectric Constant (DK) Dissipation Factor (DF)					
					100 MHz	500 MHz	1 GHz	2 GHz	5 GHz	10 GHz
1x106	71.0%	Standard	0.0020	0.051	3.89 0.0180	3.84 0.0210	3.81 0.0240	3.77 0.0250	3.63 0.0300	3.63 0.0300
1x1067	65.0%	Alternate	0.0020	0.051	3.99 0.0170	3.94 0.0200	3.91 0.0220	3.88 0.0230	3.74 0.0280	3.74 0.0280
1x1080	58.0%	Standard	0.0025	0.064	4.11 0.0160	4.06 0.0180	4.04 0.0210	4.00 0.0220	3.88 0.0260	3.88 0.0260
1x1080	59.0%	Standard	0.0030	0.076	4.09 0.0160	4.04 0.0180	4.02 0.0210	3.99 0.0220	3.86 0.0260	3.86 0.0260
1x2113	46.0%	Alternate	0.0030	0.076	4.34 0.0140	4.30 0.0160	4.28 0.0180	4.25 0.0190	4.14 0.0220	4.14 0.0220
1x1086	60.0%	Alternate	0.0030	0.076	4.07 0.0160	4.02 0.0190	4.00 0.0210	3.97 0.0220	3.84 0.0260	3.84 0.0260

High performance HF-materials:

(these are not FR-4 materials)
constant D_k / D_f values over a broad range of frequencies



e.g. AstraMT77

High Frequency material, high performance resin system

Core Data

Standard/ Alternate	Thickness (inch)	Thickness (mm)	Dielectric Constant (DK) Dissipation Factor (DF)				
			2 GHz	5 GHz	10 GHz	15 GHz	20 GHz
Standard	0.0025	0.064	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017
Standard	0.0050	0.127	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017
Standard	0.0075	0.191	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017
Standard	0.0100	0.254	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017
Standard	0.0125	0.318	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017	3.00 0.0017

4. Base Material Characteristics

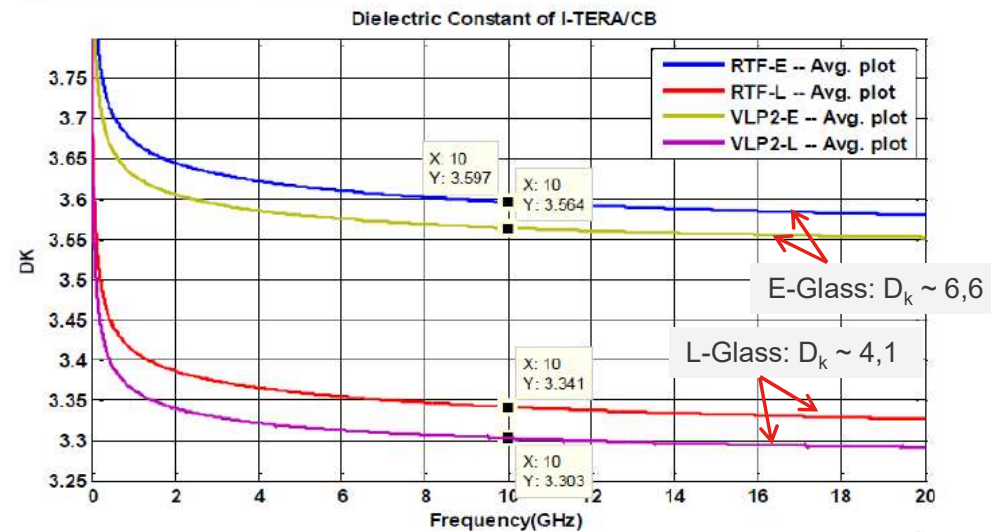
Glass

Glass has a D_k value as well as the resin.

- E-Glass $D_k \sim 6.6$
- Base material D_k depends on the resin/glass ratio

Glass fabric D_k value on the glass fabric

Comparison for RTF-E, RTF-L, VLP2-E, VLP2-L



Material: I-TeraMT40

RTF: Reverse Treated Foil/ $R_z \leq 5 \mu m$

VLP2: Very Low Profile Foil/ $R_z \sim 2 \mu m$

E: Standard E-Glass

L: Low-Dk-Glass

4. Base Material Characteristics

Glass



Glass style impact on base materials D_k value and impedance.

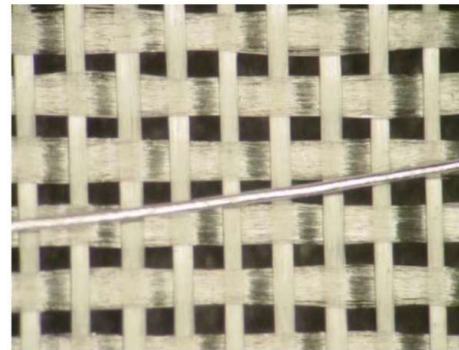


Figure 1. 1080 Glass Cloth with 3.5 Mil Wire

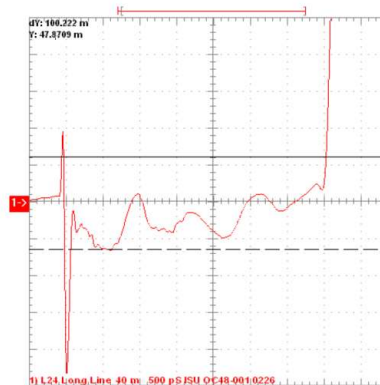


Figure 2. Impedance vs. Length Over 1080 Glass

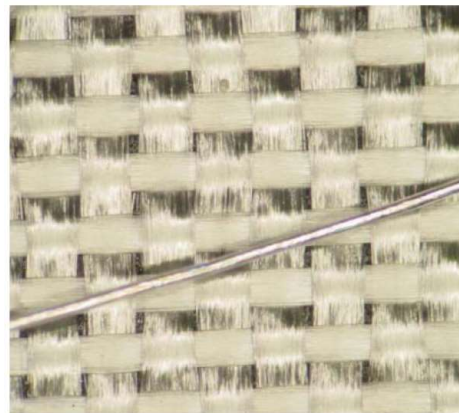


Figure 3. 3313 Glass Cloth with 3.5 Mil Wire

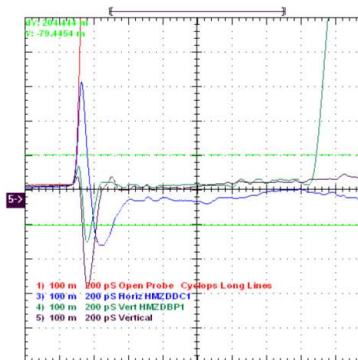
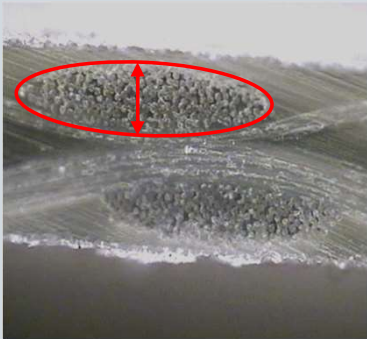


Figure 4. Impedance vs. Length Over 3313 Glass

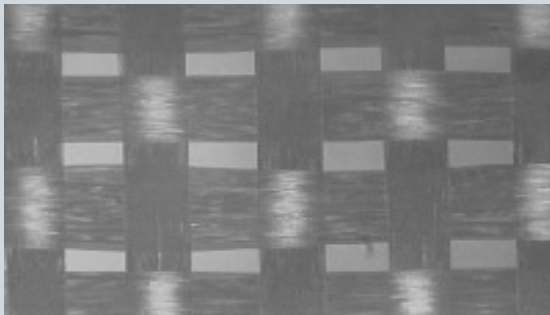
“By simply changing the style of glass used in the laminate, the problems of varying impedance and velocity have been substantially reduced.”

4. Base Material Characteristics

Glass: Spreaded Glass



1067 Standard

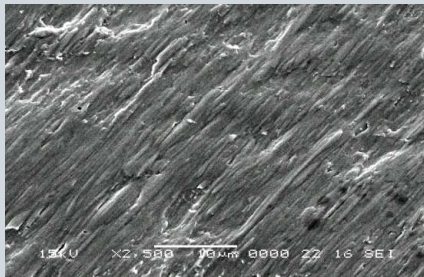


4. Base Material Characteristics

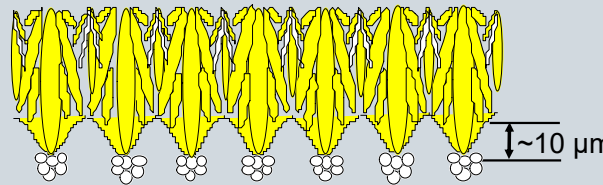
Copper



Resist side



Standard foil

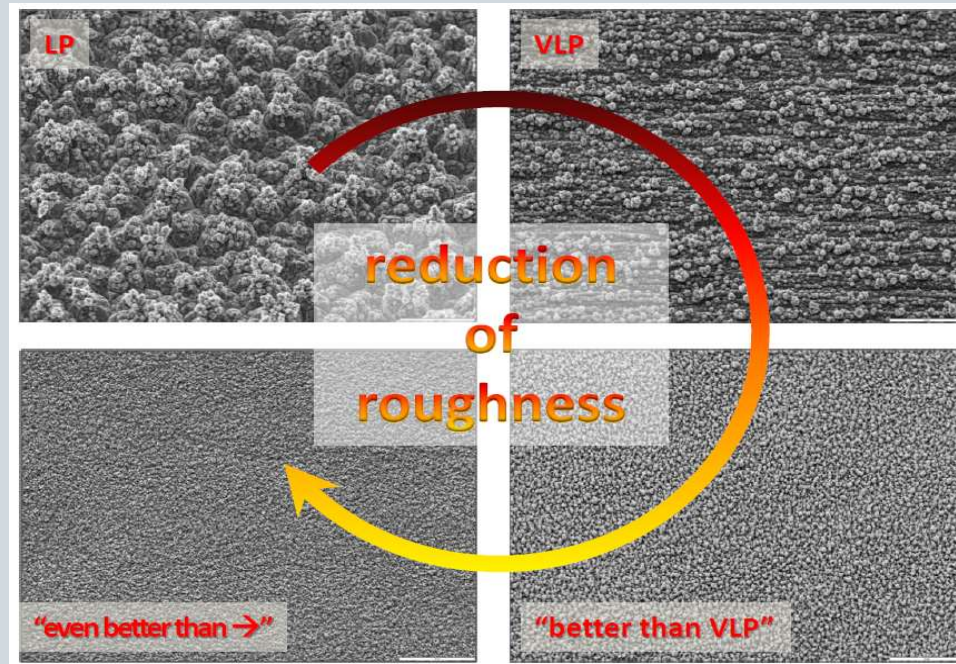


Bonding side



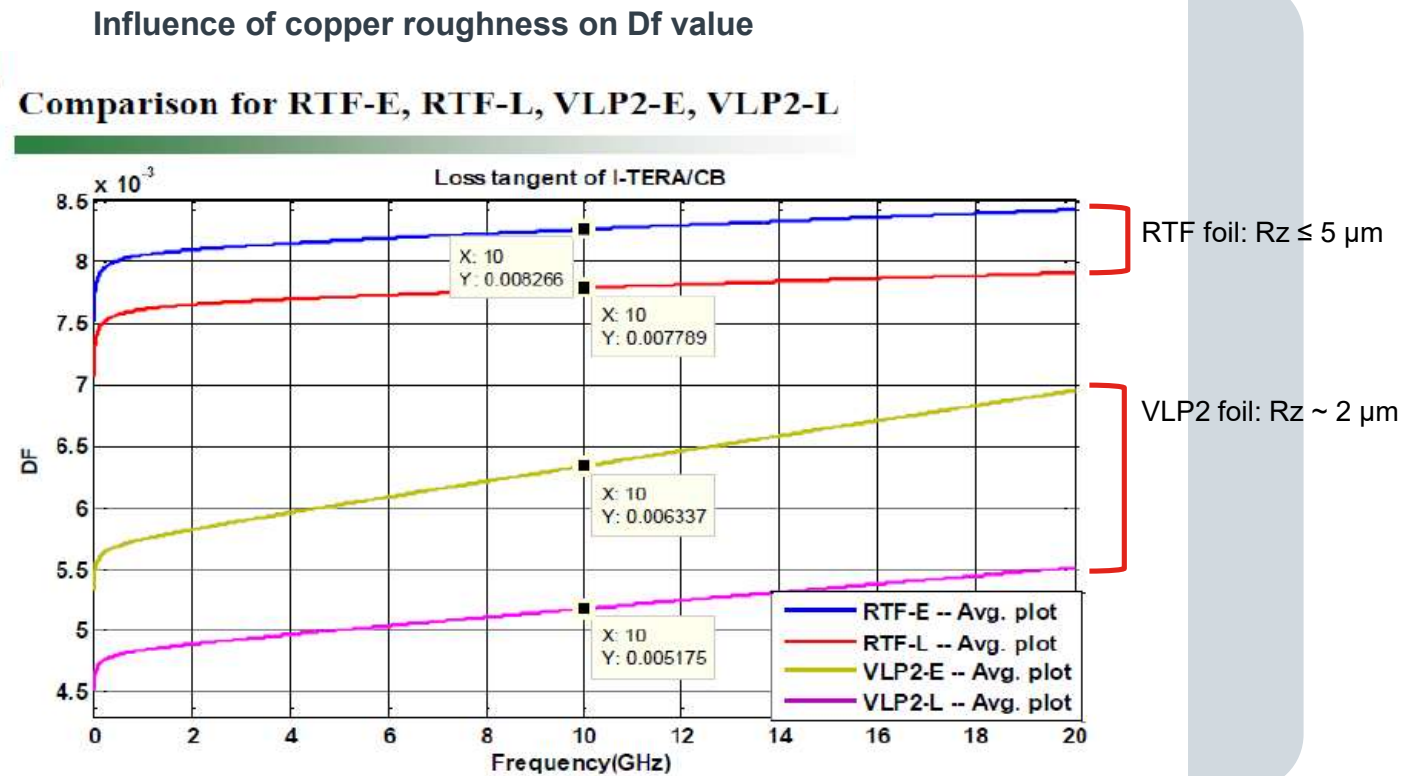
4. Base Material Characteristics

Copper



4. Base Material Characteristics

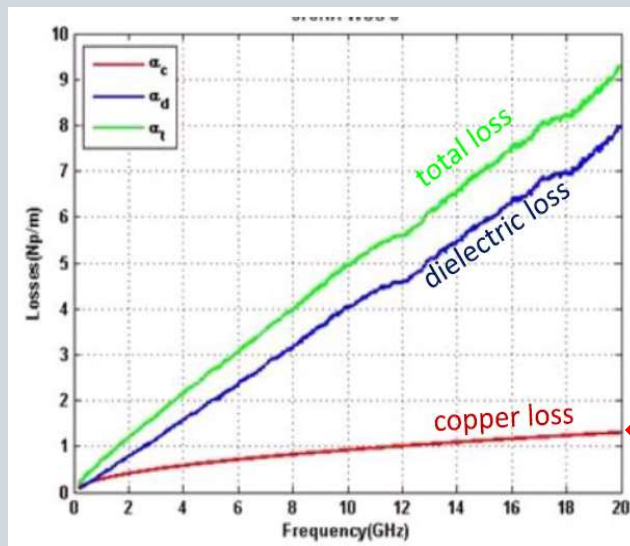
Copper



Material: I-TeraMT40
RTF: Reverse Treated Foil/ Rz ≤ 5 μm
VLP2: Very Low Profile Foil/ Rz ~ 2 μm
E: Standard E-Glas
L: Low-Dk-Glas

4. Base Material Characteristics

Signal Attenuation: influence of resin and copper



D_f value = resin

Copper treatment roughness: Skin Effect

5. Specifications and Standards

5. Specifications and Standards

IPC: **Institute of Printed Circuits**

IPC is a trade association whose aim is to standardize the assembly and production requirements of electronic equipment and assemblies. (Wikipedia)

Main IPC's for Base Materials:

- ✧ IPC-4562 Metal Foil for Printed Wirings Applications
- ✧ IPC-4412 Specification for Finished Fabric Woven from „E“ Glass for Printed Boards
- ✧ **IPC-4101 Specification for Base Materials for Rigid und Multilayer Printed Boards**
- ✧ **IPC-TM-650 Test Method Manual**

UL: **Underwriters Laboratories**

UL is a global safety certification company.

5. Specifications and Standards: IPC

IPC-4101: Specification for Base Materials for Rigid und Multilayer Printed Boards



Determination of basics like:

- › defines glass fabric type / copper / resin
- › Class of surface qualities
- › Definition of Length/width and thickness of laminates

Specified definition of base material characteristics (to a certain extent):

- › Defines physical / chemical / electrical requirements of prepregs and laminates e.g. Peel strength, Dimensional Stability, Tg, ...
- › IPC categorizes materials according to the resin system and material properties in the specification sheets ("**Slash Sheets**")
- › Base Materials are delivered according to IPC-4101 slash sheets

5. Specifications and Standards: IPC

IPC-4101: Specification for Base Materials for Rigid und Multilayer Printed Boards



How is a base material assigned to a slash sheet?

- The base material producer classifies its materials according to
 - Resin type
 - Material characteristics

5. Specifications and Standards: IPC

IPC-4101: Specification for Base Materials for Rigid und Multilayer Printed Boards

Slash Sheet categories:



- › Type of base material: e.g. FR-4
- › Filled/unfilled system
- › Halogen containing or halogen free system

example /21: basic slash sheet for FR-4 materials

/21	FR-4.0 GFN Epoxy / Woven Glass
-----	--------------------------------------

SPECIFICATION SHEET #: IPC-4101/21			Keywords: (For Search Only) NOT Grade Requirement See Section 7
REINFORCEMENT: 1: Woven E-glass		2: NONE	
RESIN SYSTEM: <i>Primary:</i> Difunctional epoxy		<i>Secondary 2:</i> NONE	FR-4.0
	<i>Secondary 1:</i> Multifunctional epoxy	<i>Minimum UL94 Requirement:</i> V-0	GFN
FLAME RETARDANT MECHANISM: RoHS Compliant Bromine			Epoxy / Woven Glass
FILLERS: <5%			
ID REFERENCE: <i>UL/ANSI:</i> FR-4.0/21		<i>MIL-S-13949:</i> /04	
GLASS TRANSITION (T_g): 110 °C minimum		- GF, GFN, GFK, GFP, GFM	

5. Specifications and Standards: IPC



Slash Sheet /21

IPC-4101E

March 2017

Revision Date: March 2017

SPECIFICATION SHEET

SPECIFICATION SHEET #: IPC-4101/21

REINFORCEMENT: 1: Woven E-glass

RESIN SYSTEM: Primary: Difunctional epoxy
Secondary 1: Multifunctional epoxy
Secondary 2: NONE

FLAME RETARDANT MECHANISM: RoHS Compliant Bromine

FILLERS: <5%

ID REFERENCE: UL/ANSI: FR-4.0/21

GLASS TRANSITION (T_g): 110 °C minimum

Z: NONE

Minimum UL94 Requirement: V-0

MIL-S-13949: /04
-GF, GFN, GFK, GFP, GFM

LAMINATE REQUIREMENTS

Laminate Requirement	Specification <0.50 mm [0.0197 in]	Specification >0.50 mm [0.0197 in]	Units	Test Method	Ref. Para.
1. Peel Strength, minimum					3.9.1.1
A. Low profile copper foil and very low profile copper foil – all copper foil >17 µm [0.669 mil]	0.70 [4.00]	0.70 [4.00]	N/mm [lb/in]	2.4.6	3.9.1.1.1 3.9.1.1.2 3.9.1.1.3
B. Standard profile copper foil				2.4.8.2	
1. After thermal stress	0.80 [4.57]	1.05 [6.00]		2.4.8.3	
2. At 125 °C [257 °F]	0.70 [4.00]	0.70 [4.00]			
3. After process solutions	0.55 [3.14]	0.80 [4.57]			
C. All other foil – composite	AABUS	AABUS			
2. Volume Resistivity, minimum					
A. 96/35/90	10 ⁹	–	MΛ-cm	2.5.17.1	3.11.1.3
B. After moisture resistance	–	10 ⁹			
C. At elevated temperature E-24/125	10 ⁹	10 ⁹			
3. Surface Resistivity, minimum					
A. 96/35/90	10 ⁴	–	MΛ	2.5.17.1	3.11.1.4
B. After moisture resistance	–	10 ⁴			
C. At elevated temperature E-24/125	10 ³	10 ³			
4. Moisture Absorption, maximum	–	0.80	%	2.6.2.1	3.12.1.1
5. Dielectric Breakdown, minimum	–	40	kV	2.5.6	3.11.1.6
6. Permittivity at 1 MHz, maximum (Laminate & laminated prepreg)*	5.4	5.4	–	2.5.5.2 2.5.5.3 2.5.5.9	3.11.1.1 3.11.2.1
7. Loss Tangent at 1 MHz, maximum (Laminate & laminated prepreg)*	0.035	0.035	–	2.5.5.2 2.5.5.3 2.5.5.9	3.11.1.2 3.11.2.2
8. Flexural Strength, minimum					
A. Length direction	–	415 [60,190]	N/mm ² [lb/in ²]	2.4.4	3.9.1.3
B. Cross direction	–	345 [50,040]			
9. Flexural Strength at Elevated Temperature, length direction, minimum	–	–	N/mm ² [lb/in ²]	2.4.4.1	3.9.1.4
10. Arc Resistance, minimum	60	60	s	2.5.1	3.11.1.5
11. Thermal Stress 10 s at 286 °C [550.4 °F], minimum					
A. Unetched	Pass Visual	Pass Visual	rating	2.4.13.1	3.10.1.2
B. Etched	Pass Visual	Pass Visual			
12. Electric Strength, minimum (Laminate & laminated prepreg)	30	–	kV/mm	2.5.5.2	3.11.1.7 3.11.2.3
13. Flammability, minimum (Laminate & laminated prepreg)	V-0	V-0	rating	UL94	3.10.2.1 3.10.1.1
14. Glass Transition Temperature, minimum	–	110	°C	2.4.24 2.4.25	3.10.1.6
15. Other	–	–			

*See Slash Sheet 2 in IPC-4121.

*See Slash Sheet 2 in IPC-4121.

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PREPREG REQUIREMENTS				
Prepreg Requirement	Specification	Units	Test Method	Ref. Para.
1. Shelf Life, minimum (Condition 1/Condition 2)	180/90	Days	AABUS	3.17
2. Reinforcement	As per IPC-4412 or AABUS.			
3. Volatile content maximum	0.75	%	2.3.19	3.9.2.2.8
4. Prepreg Parameters	–	AABUS	AABUS	1.1.7
5. Flammability, minimum (as laminated)	V-0	rating	UL94	3.10.2.1
6. Other	–			



Slash Sheet limitations

- min/max values are defined that are not very specific
- A high slash sheet number is not a guarantee for every aspect of high performance

5. Specifications and Standards: IPC



Slash Sheet /21

IPC-4101E

March 2017

Revision Date: March 2017

SPECIFICATION SHEET

SPECIFICATION SHEET #: IPC-4101/21

REINFORCEMENT: 1: Woven E-glass

RESIN SYSTEM: Primary: Difunctional epoxy
Secondary 1: Multifunctional epoxy
Secondary 2: NONE

FLAME RETARDANT MECHANISM: RoHS Compliant Bromine

FILLERS: <5%

ID REFERENCE: UL/ANSI: FR-4.0/21

GLASS TRANSITION (T_g): 110 °C minimum

2: NONE

Minimum UL94 Requirement: V-0

MIL-S-13949: /04
-GF, GFN, GFK, GFP, GFM

LAMINATE REQUIREMENTS

Laminate Requirement	Specification <0.50 mm [0.0197 in]	Specification >0.50 mm [0.0197 in]	Units	Test Method	Ref. Para.
1. Peel Strength, minimum					3.9.1.1
A. Low profile copper foil and very low profile copper foil – all copper foil >17 µm [0.669 mil]	0.70 [4.00]	0.70 [4.00]	N/mm [lb/in]	2.4.6	3.9.1.1.1 3.9.1.1.2 3.9.1.1.3
B. Standard profile copper foil				2.4.8.2	
1. After thermal stress	0.80 [4.57]	1.05 [6.00]		2.4.8.3	
2. At 125 °C [257 °F]	0.70 [4.00]	0.70 [4.00]			
3. After process solutions	0.55 [3.14]	0.80 [4.57]			
C. All other foil – composite	AABUS	AABUS			
2. Volume Resistivity, minimum					
A. 96/35/90	10 ⁹	–	MΛ-cm	2.5.17.1	3.11.1.3
B. After moisture resistance	–	10 ⁹			
C. At elevated temperature E-24/125	10 ⁹	10 ⁹			
3. Surface Resistivity, minimum					
A. 96/35/90	10 ⁴	–	MΛ	2.5.17.1	3.11.1.4
B. After moisture resistance	–	10 ⁴			
C. At elevated temperature E-24/125	10 ³	10 ³			
4. Moisture Absorption, maximum	–	0.80	%	2.6.2.1	3.12.1.1
5. Dielectric Breakdown, minimum	–	40	kV	2.5.6	3.11.1.6
6. Permittivity at 1 MHz, maximum (Laminate & laminated prepreg)*	5.4	5.4	–	2.5.5.2 2.5.5.3 2.5.5.9	3.11.1.1 3.11.2.1
7. Loss Tangent at 1 MHz, maximum (Laminate & laminated prepreg)*	0.035	0.035	–	2.5.5.2 2.5.5.3 2.5.5.9	3.11.1.2 3.11.2.2
8. Flexural Strength, minimum					
A. Length direction	–	415 [60,190]	N/mm ² [lb/in ²]	2.4.4	3.9.1.3
B. Cross direction	–	345 [50,040]			
9. Flexural Strength at Elevated Temperature, length direction, minimum	–	–	N/mm ² [lb/in ²]	2.4.4.1	3.9.1.4
10. Arc Resistance, minimum	60	60	s	2.5.1	3.11.1.5
11. Thermal Stress 10 s at 286 °C [550.4 °F], minimum					
A. Unetched	Pass Visual	Pass Visual	rating	2.4.13.1	3.10.1.2
B. Etched	Pass Visual	Pass Visual			
12. Electric Strength, minimum (Laminate & laminated prepreg)	30	–	kV/mm	2.5.5.2	3.11.1.1 3.11.2.3
13. Flammability, minimum (Laminate & laminated prepreg)	V-0	V-0	rating	UL94	3.10.2.1 3.10.1.1
14. Glass Transition Temperature, minimum	–	110	°C	2.4.24 2.4.25	3.10.1.6
15. Other	–	–			

*See Slash Sheet 2 in IPC-4101.

*See Slash Sheet 2 in IPC-4121.

March 2017				
IPC-4101E				
PREPREG REQUIREMENTS				
Prepreg Requirement	Specification	Units	Test Method	Ref. Para.
1. Shelf Life, minimum (Condition 1/Condition 2)	180/90	Days	AABUS	3.17
2. Reinforcement	As per IPC-4412 or AABUS.			
3. Volatile content maximum	0.75	%	2.3.19	3.9.2.2.8
4. Prepreg Parameters	–	AABUS	AABUS	1.1.7
5. Flammability, minimum (as laminated)	V-0	rating	UL94	3.10.2.1
6. Other	–			

Test methods are usually described in the IPC-TM-650 “Test Methods Manual”

→ “Numbers without letters” always refers to IPC-TM-650

An example of an exception is the UL flammability test.

5. Specifications and Standards: UL

Underwriters Laboratories (UL)



- › Independent organisation for the investigation of product safety, especially electrical safety and flame retardancy (flammability)
- › many UL standards are part of ANSI standards (American National Standards Institute)
- › the UL marking is not required by law, unlike the CE marking, but has a similar weight in North America

5. Specifications and Standards: UL

Relevant standards for Polymeric Materials of the UL 746 series:

- **UL 746A Short Term Property Evaluations**
 - Wide range of individual tests, e.g. Flammability, IR
- **UL 746B Long Term Property Evaluations**
 - Thermal aging at four elevated temperatures
 - Minimum duration of time: 5.000 hours
- **UL 746E Industrial Laminates, Filament Wound Tubing, Vulcanized Fibre, and Materials Used in Printed-Wiring Boards**
 - Bond strength delamination and blistering test
- All Base Material Suppliers have an own “UL File”, where all base materials are listed with the needed and UL proved properties.
- **FR-4 became FR-4.0 and FR-4.1**
 - FR-4.0: brominated epoxy materials
 - FR-4.1: low halogen content epoxy materials



5. Specifications and Standards: UL

UL 746A: Short Term Property Evaluations

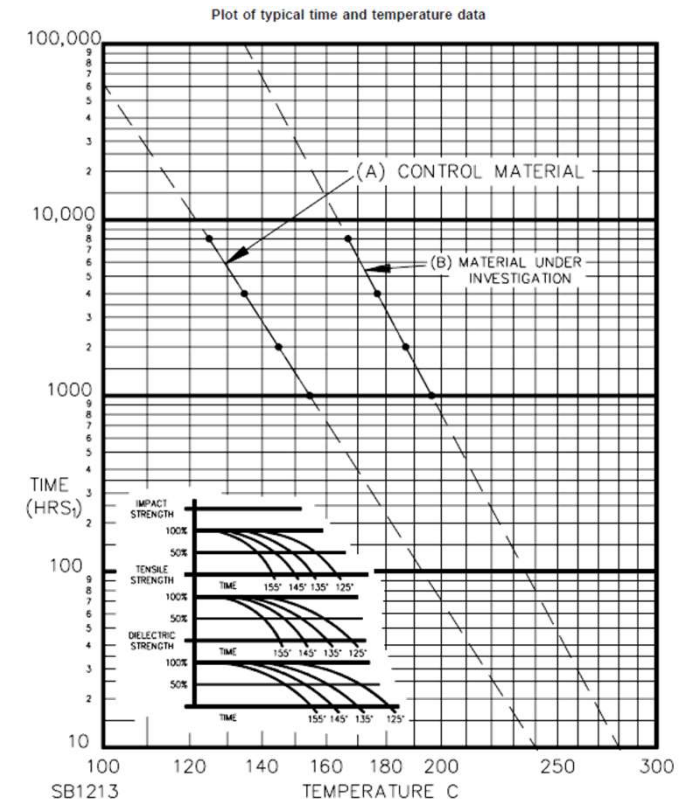


- › Comparative Tracking Index (CTI) test
- › Flammability
- › Electrical properties
- › Mechanical properties
- › IR spectrum

5. Specifications and Standards: UL

UL 746B: Long Term Property Evaluations

- Definition of a temperature at which critical material properties remain within an acceptable range after e.g. 60,000 hours
- "Acceptable range": loss of characteristic max. 50%
- Ageing test for **base material laminates**
- Testing results give the RTI (Relative Temperature Index). It's the maximum temperature at which a material will maintain acceptable properties within a specific time.



5. Specifications and Standards: UL

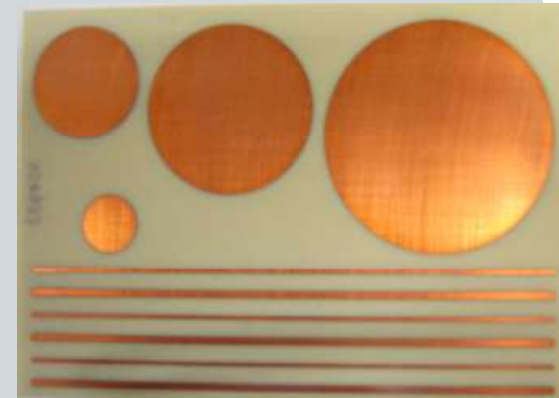
UL 746E: Industrial Laminates, Filament Wound Tubing, Vulcanized Fibre, and Materials Used in Printed-Wiring Boards

- Test with copper-clad multilayer materials
- Storage of samples at defined temperatures
- Testing results give the MOT (Maximum Operating Temperature)
MOT can not be higher than RTI
MOT is not the Continuous Operating Temperature

- FR-4 materials: max defined MOT of 130 °C
FR-5 materials: max defined MOT of 150 °C

- Two test conditions are possible:
 - MOT 10 days test:
 $T = 1.076 (T_{\text{MOT}} + 288) - 273$
or
 - MOT 56 days test:
 $T = 1.02 (T_{\text{MOT}} + 288) - 273$

- Test criteria:
 - no delaminations
 - copper peel strength $\geq 0,175 \text{ N/mm}$



UL 746E Layout

	10 Tage	56 Tage
MOT 130	177 °C	153 °C
MOT 150	198 °C	174 °C

ALMOST DONE....

PLEASE DO NOT FORGET...

- The pcb manufacturer is the expert for pcb´s!
- The base material supplier is the expert for base materials!
- The pcb manufacturer is closest to the base material and can choose the right material having the right information.
- Let the pcb shop choose the prepreg/core type, specify only the final thickness and isolation thickness as often as possible.
- Specify as much as necessary and as little as possible!
- If you are not sure about something, ask your suppliers!

Thank You! Questions?

Adolf Sommer

Head of Technical Service and
Application Engineering Central Europe

Mobile: +49 173 2525 583

adolf.sommer@isola-group.com

isola[®]